

Complex Geometry

Day 1

These notes develop the main definitions and arguments in detail, including clarifications of potentially confusing points.

1 One complex variable and the $\bar{\partial}$ operator

[Definition] Let $z = x + iy$. The Wirtinger derivatives are

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

They are normalized by

$$\frac{\partial z}{\partial z} = 1, \quad \frac{\partial \bar{z}}{\partial z} = 0, \quad \frac{\partial z}{\partial \bar{z}} = 0, \quad \frac{\partial \bar{z}}{\partial \bar{z}} = 1.$$

For a smooth function f ,

$$df = \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z} = \partial f + \bar{\partial} f.$$

Thus ∂f has type $(1, 0)$ and $\bar{\partial} f$ has type $(0, 1)$. The holomorphicity criterion is

$$f \in \mathcal{O}(\Omega) \iff \bar{\partial} f = 0.$$

The decomposition $d = \partial + \bar{\partial}$ is the complex analogue of resolving the real differential into components compatible with the complex structure.

2 Cauchy-Green formula

[Theorem] For a bounded domain $\Omega \subset \mathbb{C}$ with sufficiently regular boundary and $f \in C^1(\bar{\Omega})$, the Cauchy-Green formula is

$$f(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{f(w)}{w - z} dw + \frac{1}{2\pi i} \int_{\Omega} \frac{\partial f / \partial \bar{w}}{w - z} dw \wedge d\bar{w}.$$

The first term is the boundary Cauchy integral; the second measures the failure of f to be holomorphic. If $\bar{\partial} f = 0$, the second term vanishes.

3 Solving $\bar{\partial}u = f d\bar{z}$ on $\mathbb{C}$

[Definition] Let $f \in C_c^\infty(\mathbb{C})$. The Cauchy transform is

$$Tf(z) = \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{f(w)}{w - z} dw \wedge d\bar{w}.$$

[Lemma] This transform satisfies

$$\frac{\partial}{\partial \bar{z}} Tf(z) = f(z),$$

so $u = Tf$ solves

$$\bar{\partial}u = f d\bar{z}.$$

Why the singular integral is meaningful

The kernel $|w - z|^{-1}$ is locally integrable in two real dimensions:

$$\int_{|w-z|<\varepsilon} \frac{1}{|w-z|} dA(w) = 2\pi\varepsilon < \infty.$$

Compact support controls the integral at infinity. Thus the displayed integral is absolutely meaningful for smooth compactly supported f .

Differentiation by translation

Writing $w = z + \zeta$ gives

$$Tf(z) = \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{f(z + \zeta)}{\zeta} d\zeta \wedge d\bar{\zeta}.$$

The domain of integration is now independent of z . Formally differentiating gives

$$\frac{\partial Tf}{\partial \bar{z}}(z) = \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{\partial f}{\partial \bar{z}}(z + \zeta) \frac{d\zeta \wedge d\bar{\zeta}}{\zeta}.$$

Applying the Cauchy-Green/Stokes formula to a large disk containing the support yields $f(z)$.

Clarification about compact support

[Remark] Compact support is used twice. First, it makes the integral converge at infinity. Second, it permits integration by parts/Stokes on a large disk with no outer-boundary contribution, because the integrand vanishes near that boundary. Local integrability handles the pole at $w = z$; compact support handles infinity and the boundary term.

4 Holomorphic functions of several variables

Let $\Omega \subset \mathbb{C}^n$ be open, with coordinates

$$z = (z_1, \dots, z_n), \quad z_j = x_j + iy_j,$$

and norm $|z|^2 = \sum_{j=1}^n |z_j|^2$. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$, write

$$|\alpha| = \alpha_1 + \dots + \alpha_n, \quad \alpha! = \alpha_1! \cdots \alpha_n!, \quad z^\alpha = z_1^{\alpha_1} \cdots z_n^{\alpha_n}.$$

[Notation] The Euclidean ball and polydisc are

$$B(a, r) = \{z \in \mathbb{C}^n : |z - a| < r\},$$

$$\Delta(a; r_1, \dots, r_n) = \prod_{j=1}^n \Delta(a_j, r_j).$$

[Definition] A C^1 function $f : \Omega \rightarrow \mathbb{C}$ is holomorphic if and only if

$$\frac{\partial f}{\partial \bar{z}_j} = 0, \quad j = 1, \dots, n,$$

or equivalently $\bar{\partial}f = 0$, where

$$\bar{\partial}f = \sum_{j=1}^n \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j.$$

Cauchy formula on a polydisc

[Theorem] If f is holomorphic on a neighbourhood of the closed polydisc, repeated use of the one-variable Cauchy formula gives

$$f(z) = \frac{1}{(2\pi i)^n} \int_{|w_1 - a_1| = r_1} \cdots \int_{|w_n - a_n| = r_n} \frac{f(w_1, \dots, w_n)}{\prod_{j=1}^n (w_j - z_j)} dw_1 \cdots dw_n.$$

This is not a new independent formula: it follows by holding $n - 1$ variables fixed, applying the one-variable theorem in the remaining variable, and iterating.

Consequences include:

- the maximum-modulus principle;
- Cauchy estimates for every mixed derivative;
- a convergent Taylor expansion in multi-index form,

$$f(z) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^n} \frac{1}{\alpha!} \partial^\alpha f(a) (z - a)^\alpha;$$

- the identity principle on a connected domain.

[Remark] The several-variable identity principle must be interpreted correctly: agreement on a nonempty open subset implies agreement everywhere on the connected domain. Agreement merely on an arbitrary set with an accumulation point is not sufficient in $\mathbb{C}^n$ when $n \geq 2$, because a nonzero holomorphic function may vanish on a hypersurface.

5 Differential forms of type (p, q)

[Definition] Every complex-valued one-form can be written uniquely as a sum of types:

$$\gamma = \sum_{j=1}^n a_j dz_j + \sum_{j=1}^n b_j d\bar{z}_j = \gamma^{1,0} + \gamma^{0,1}.$$

More generally, a (p, q) -form is a finite sum

$$\alpha = \sum_{|I|=p, |J|=q} \alpha_{I\bar{J}} dz_I \wedge d\bar{z}_J.$$

[Remark] The pair (p, q) here records the complex bidegree: p is the number of dz factors and q is the number of $d\bar{z}$ factors. It is not the same notion as the type (s, r) of a tensor, where the two entries count contravariant and covariant slots. The ordered-pair notation is simply being overloaded.

The exterior derivative splits as

$$d = \partial + \bar{\partial},$$

with

$$\partial : \Omega^{p,q} \rightarrow \Omega^{p+1,q}, \quad \bar{\partial} : \Omega^{p,q} \rightarrow \Omega^{p,q+1}.$$

Since $d^2 = 0$ and the three resulting terms have different bidegrees,

$$\partial^2 = 0, \quad \bar{\partial}^2 = 0, \quad \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

For a $(0, 1)$ -form

$$\alpha = \sum_{j=1}^n a_j d\bar{z}_j,$$

one computes

$$\bar{\partial}\alpha = \sum_{j < k} \left(\frac{\partial a_k}{\partial \bar{z}_j} - \frac{\partial a_j}{\partial \bar{z}_k} \right) d\bar{z}_j \wedge d\bar{z}_k.$$

Hence $\bar{\partial}\alpha = 0$ exactly when the compatibility conditions

$$\frac{\partial a_k}{\partial \bar{z}_j} = \frac{\partial a_j}{\partial \bar{z}_k} \quad (j, k = 1, \dots, n)$$

hold. These conditions are necessary for $\alpha = \bar{\partial}u$, because $\bar{\partial}^2 u = 0$.

6 Hartogs extension theorem

Statement

[Theorem] Let $\Omega \subset \mathbb{C}^n$ with $n \geq 2$, and let $K \Subset \Omega$ be compact. Assume $\Omega \setminus K$ is connected. Every

$$f \in \mathcal{O}(\Omega \setminus K)$$

extends uniquely to a holomorphic function $\tilde{f} \in \mathcal{O}(\Omega)$.

Thus isolated compact “holes” do not produce genuine singularities in several complex variables. This is sharply different from one variable, where $1/z$ on a punctured disc cannot be extended across the origin.

Cutoff reduction

[Proof] Choose nested open sets

$$K \Subset \Omega_1 \Subset \Omega_2 \Subset \Omega$$

and a smooth cutoff $\eta \in C_c^\infty(\Omega)$ with

$$\eta = 0 \text{ near } K, \quad \eta = 1 \text{ on } \Omega \setminus \Omega_2.$$

Because f is defined on $\Omega \setminus K$, the product ηf extends smoothly across K by setting it equal to 0 near K . Its failure to be holomorphic is

$$\alpha := \bar{\partial}(\eta f) = f \bar{\partial} \eta,$$

because $\bar{\partial} f = 0$ where f is defined. Moreover, α has compact support in the transition annulus $\Omega_2 \setminus \overline{\Omega_1}$ and

$$\bar{\partial} \alpha = \bar{\partial}^2(\eta f) = 0.$$

Suppose we can solve

$$\bar{\partial} G = \alpha$$

with G compactly supported. Then

$$F := \eta f - G$$

satisfies $\bar{\partial} F = 0$, hence is holomorphic on all of Ω . Outside a compact set, both $\eta = 1$ and $G = 0$, so $F = f$ on a nonempty open subset of $\Omega \setminus K$. By the identity principle and connectedness, $F = f$ throughout $\Omega \setminus K$. This F is the desired extension.

The proof is therefore reduced to a compact-support $\bar{\partial}$ lemma.

Compact-support $\bar{\partial}$ lemma

[Lemma] Let $n \geq 2$ and

$$\alpha = \sum_{j=1}^n a_j d\bar{z}_j$$

be a smooth, compactly supported $(0,1)$ -form on $\mathbb{C}^n$ satisfying $\bar{\partial}\alpha = 0$. Then there exists $G \in C_c^\infty(\mathbb{C}^n)$ such that

$$\bar{\partial}G = \alpha.$$

[Proof] Start by solving in the first variable with all other variables as parameters:

$$G(z) = \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{a_1(w, z_2, \dots, z_n)}{w - z_1} dw \wedge d\bar{w}.$$

The one-variable formula gives

$$\frac{\partial G}{\partial \bar{z}_1} = a_1.$$

For $j \geq 2$, differentiate under the integral and use $\bar{\partial}\alpha = 0$:

$$\begin{aligned} \frac{\partial G}{\partial \bar{z}_j} &= \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{\partial a_1 / \partial \bar{z}_j}{w - z_1} dw \wedge d\bar{w} \\ &= \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{\partial a_j / \partial \bar{w}}{w - z_1} dw \wedge d\bar{w} \\ &= a_j(z). \end{aligned}$$

The final equality is the one-variable Cauchy-Green formula (or Stokes' theorem). Therefore $\bar{\partial}G = \alpha$.

Why G has compact support when $n \geq 2$

The first integral formula immediately shows that G vanishes when the transverse variables $(z_2, \dots, z_n)$ lie outside the projection of $\text{supp } \alpha$. It does not yet control the z_1 direction. Use the second complex variable and define

$$\tilde{G}(z) = \frac{1}{2\pi i} \int_{\mathbb{C}} \frac{a_2(z_1, w, z_3, \dots, z_n)}{w - z_2} dw \wedge d\bar{w}.$$

The same calculation, now performed in the z_2 variable, gives

$$\bar{\partial}\tilde{G} = \alpha.$$

Thus $G - \tilde{G}$ is holomorphic on $\mathbb{C}^n$. Choose $R > 0$ such that

$$\text{supp } \alpha \subset \{z \in \mathbb{C}^n : |z_j| < R \text{ for every } j\}.$$

On the nonempty open set

$$\{z : |z_1| > R, |z_2| > R\},$$

the first representation gives $G = 0$, since $|z_2| > R$, and the second gives $\tilde{G} = 0$, since $|z_1| > R$. Hence $G - \tilde{G}$ vanishes on a nonempty open set. By the identity principle,

$$G = \tilde{G} \quad \text{on } \mathbb{C}^n.$$

We can now use both representations. The first gives $G = 0$ whenever one of $z_2, \dots, z_n$ lies outside the corresponding projection of $\text{supp } \alpha$. Since $G = \tilde{G}$, the second gives $G = 0$ whenever one of $z_1, z_3, \dots, z_n$ lies outside the corresponding projection. Every coordinate of a point in $\text{supp } G$ is therefore bounded, so G has compact support.

This is precisely where $n \geq 2$ matters: the second variable produces another representation of the same solution, and the two representations together control every coordinate direction.

Day 2

7 Holomorphic maps and biholomorphisms

[Definition] Let $\Omega \subset \mathbb{C}^n$ and $\Omega' \subset \mathbb{C}^m$ be domains. A map

$$F = (f^1, \dots, f^m) : \Omega \longrightarrow \Omega'$$

is holomorphic if every component f^α is holomorphic. Equivalently,

$$\frac{\partial f^\alpha}{\partial \bar{z}_j} = 0, \quad 1 \leq \alpha \leq m, \quad 1 \leq j \leq n.$$

Its complex differential at $a \in \Omega$ is the complex-linear map represented by the complex Jacobian

$$D_{\mathbb{C}}F(a) = J_{\mathbb{C}}F(a) = \left(\frac{\partial f^\alpha}{\partial z_j}(a) \right)_{\alpha, j}.$$

Thus

$$F(a + h) = F(a) + D_{\mathbb{C}}F(a)h + o(|h|).$$

In particular, a map from a domain in $\mathbb{C}$ to $\mathbb{C}^n$ is holomorphic exactly when each of its n component functions is holomorphic.

A map $F : \Omega \rightarrow \Omega'$ is biholomorphic if it is bijective, holomorphic, and its inverse $F^{-1} : \Omega' \rightarrow \Omega$ is holomorphic. In that case Ω and Ω' are said to be biholomorphic.

8 Complex and real Jacobians

[Lemma] Let $F : \Omega \subset \mathbb{C}^n \rightarrow \mathbb{C}^n$ be holomorphic. Under the identification $\mathbb{C}^n \cong \mathbb{R}^{2n}$,

$$\det_{\mathbb{R}} J_{\mathbb{R}} F(a) = \left| \det_{\mathbb{C}} J_{\mathbb{C}} F(a) \right|^2.$$

In particular,

$$J_{\mathbb{R}} F(a) \text{ is invertible} \iff J_{\mathbb{C}} F(a) \text{ is invertible.}$$

[Proof] Write

$$F = u + iv, \quad z = x + iy,$$

where $u, v : \Omega \rightarrow \mathbb{R}^n$, and set

$$A = \frac{\partial u}{\partial x}, \quad B = \frac{\partial v}{\partial x}.$$

The Cauchy-Riemann equations give

$$\frac{\partial u}{\partial y} = -B, \quad \frac{\partial v}{\partial y} = A.$$

Ordering the real coordinates as (x, y) , we obtain

$$J_{\mathbb{R}} F = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}, \quad J_{\mathbb{C}} F = A + iB.$$

Complex row and column operations give

$$\begin{pmatrix} I & iI \\ I & -iI \end{pmatrix} \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \begin{pmatrix} \frac{1}{2}I & \frac{1}{2}I \\ -\frac{i}{2}I & \frac{i}{2}I \end{pmatrix} = \begin{pmatrix} A + iB & 0 \\ 0 & A - iB \end{pmatrix}.$$

The first and third matrices on the left-hand side are inverses. Taking determinants gives

$$\det_{\mathbb{R}} J_{\mathbb{R}} F = \det(A + iB) \det(A - iB) = \left| \det_{\mathbb{C}} J_{\mathbb{C}} F \right|^2.$$

This proves both assertions.

Holomorphic inverse function theorem

[Theorem] Let $F : \Omega \subset \mathbb{C}^n \rightarrow \mathbb{C}^n$ be holomorphic and let $a \in \Omega$. If

$$\det J_{\mathbb{C}} F(a) \neq 0,$$

then there are neighbourhoods U of a and V of $F(a)$ such that

$$F|_U : U \longrightarrow V$$

is biholomorphic. Moreover,

$$D_{\mathbb{C}}(F|_U)^{-1}(F(z)) = (D_{\mathbb{C}}F(z))^{-1}.$$

[Proof] The determinant identity first gives

$$\det J_{\mathbb{R}}F(a) = |\det J_{\mathbb{C}}F(a)|^2 > 0.$$

The real inverse function theorem therefore produces open neighbourhoods U of a and V of $F(a)$ together with a C^1 inverse

$$G : V \longrightarrow U.$$

To see why this real inverse is automatically holomorphic, identify $\mathbb{C}^n$ with $\mathbb{R}^{2n}$ and let

$$J : \mathbb{R}^{2n} \longrightarrow \mathbb{R}^{2n}, \quad J(x, y) = (-y, x), \quad J = \begin{pmatrix} 0 & -I \\ I & 0 \end{pmatrix}.$$

Thus J represents multiplication by i . To see directly what commutation with J means, write the real derivative of a C^1 map $H = p + iq$ as

$$D_{\mathbb{R}}H = \begin{pmatrix} P & Q \\ R & S \end{pmatrix},$$

where

$$P = \frac{\partial p}{\partial x}, \quad Q = \frac{\partial p}{\partial y}, \quad R = \frac{\partial q}{\partial x}, \quad S = \frac{\partial q}{\partial y}.$$

Direct multiplication gives

$$D_{\mathbb{R}}H J = \begin{pmatrix} Q & -P \\ S & -R \end{pmatrix}, \quad J D_{\mathbb{R}}H = \begin{pmatrix} -R & -S \\ P & Q \end{pmatrix}.$$

Therefore

$$D_{\mathbb{R}}H J = J D_{\mathbb{R}}H \iff P = S, \quad Q = -R.$$

These are exactly the Cauchy-Riemann equations

$$\frac{\partial p}{\partial x} = \frac{\partial q}{\partial y}, \quad \frac{\partial p}{\partial y} = -\frac{\partial q}{\partial x}.$$

Because F is holomorphic, its real differential has this Cauchy-Riemann block form:

$$D_{\mathbb{R}}F(z) = \begin{pmatrix} A(z) & -B(z) \\ B(z) & A(z) \end{pmatrix},$$

and direct multiplication shows that

$$D_{\mathbb{R}}F(z) J = J D_{\mathbb{R}}F(z).$$

Now differentiate the identity $G \circ F = \text{id}_U$. The real chain rule gives

$$D_{\mathbb{R}}G(F(z)) D_{\mathbb{R}}F(z) = I,$$

so

$$D_{\mathbb{R}}G(F(z)) = (D_{\mathbb{R}}F(z))^{-1}.$$

The inverse matrix also commutes with J . In fact, from $AJ = JA$, multiplication on the left and right by A^{-1} gives

$$A^{-1}J = JA^{-1}.$$

Therefore

$$D_{\mathbb{R}}G(F(z)) J = J D_{\mathbb{R}}G(F(z)).$$

By the calculation above, this commutation relation is exactly the Cauchy-Riemann system for G . Since G is C^1 , it follows that G is holomorphic. Differentiating the inverse relation in complex coordinates gives

$$D_{\mathbb{C}}G(F(z)) = (D_{\mathbb{C}}F(z))^{-1}.$$

Holomorphic implicit function theorem

[Theorem] Let $\Omega \subset \mathbb{C}^n \times \mathbb{C}^m$ be open, let

$$F : \Omega \longrightarrow \mathbb{C}^m$$

be holomorphic, and suppose

$$F(a, b) = 0, \quad \det \frac{\partial F}{\partial w}(a, b) \neq 0.$$

Then there are neighbourhoods U of a and V of b and a unique holomorphic map

$$\varphi : U \longrightarrow V$$

such that

$$F(z, w) = 0 \iff w = \varphi(z)$$

for $(z, w) \in U \times V$. Its differential is

$$D\varphi(z) = - \left(\frac{\partial F}{\partial w}(z, \varphi(z)) \right)^{-1} \frac{\partial F}{\partial z}(z, \varphi(z)).$$

[Proof] Apply the holomorphic inverse function theorem to

$$\Phi(z, w) = (z, F(z, w)).$$

Its complex Jacobian is block triangular, and its lower-right block is $\partial F/\partial w$. The local holomorphic inverse of Φ gives $w = \varphi(z)$.

9 The one-variable uniformization phenomenon

[Theorem] The Riemann mapping theorem states that every nonempty simply connected proper domain

$$\Omega \subsetneq \mathbb{C}$$

is biholomorphic to the unit disc

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

Thus simply connected plane domains have a remarkably uniform biholomorphic model.

[Theorem] In several variables this fails: Poincare proved that the bidisc

$$\mathbb{D}^2 = \{(z_1, z_2) : |z_1| < 1, |z_2| < 1\}$$

is not biholomorphic to the two-dimensional unit ball

$$\mathbb{B}^2 = \{(z_1, z_2) : |z_1|^2 + |z_2|^2 < 1\}.$$

Both are bounded and simply connected, but their complex geometry is different. The classification problem is therefore already much richer in complex dimension two.

10 Domains determined by holomorphic functions

Domain of holomorphy

[Definition] A domain $\Omega \subset \mathbb{C}^n$ is a domain of holomorphy if, for every strictly larger domain

$$\Omega' \supsetneq \Omega,$$

the restriction map

$$\mathcal{O}(\Omega') \longrightarrow \mathcal{O}(\Omega), \quad F \longmapsto F|_{\Omega},$$

is not surjective. Equivalently, whenever Ω' is strictly larger than Ω , at least one holomorphic function on Ω does not extend holomorphically to Ω' .

[Lemma] Every convex domain $\Omega \subset \mathbb{C}^n$ is a domain of holomorphy.

[Proof] Fix $a \in \partial\Omega$. After translating the coordinates, one may assume that $a = 0$. Since Ω is convex, the supporting-hyperplane theorem gives a nonzero real-linear functional

$$\ell : \mathbb{C}^n \cong \mathbb{R}^{2n} \longrightarrow \mathbb{R}$$

such that

$$\ell(z) > 0 \quad \text{for every } z \in \Omega, \quad \ell(0) = 0.$$

Every real-linear functional on $\mathbb{C}^n$ can be written as

$$\ell(z) = \sum_{j=1}^n (c_j z_j + \overline{c_j} \overline{z_j}) = 2 \operatorname{Re} \left(\sum_{j=1}^n c_j z_j \right).$$

Set

$$L(z) := \sum_{j=1}^n c_j z_j.$$

The function L is complex-linear and

$$\operatorname{Re} L(z) = \frac{1}{2} \ell(z) > 0 \quad \text{on } \Omega.$$

In particular, L never vanishes on Ω , so

$$f_a(z) := \frac{1}{L(z)}$$

belongs to $\mathcal{O}(\Omega)$. As $z \rightarrow a = 0$ through Ω , one has $L(z) \rightarrow 0$, so f_a is unbounded near a . Consequently, f_a cannot extend holomorphically across a .

Finally, let $\Omega' \supsetneq \Omega$ be any larger domain. A path in Ω' from a point of Ω to a point of $\Omega' \setminus \Omega$ meets some $a \in \partial\Omega \cap \Omega'$. The corresponding function f_a cannot be the restriction of a holomorphic function on Ω' . Hence the restriction map $\mathcal{O}(\Omega') \rightarrow \mathcal{O}(\Omega)$ is not surjective, and Ω is a domain of holomorphy.

Holomorphic convexity

[Definition] For a compact set $K \Subset \Omega$, its holomorphic hull in Ω is

$$\widehat{K} = \left\{ z \in \Omega : |f(z)| \leq \sup_K |f| \text{ for every } f \in \mathcal{O}(\Omega) \right\}.$$

[Definition] The domain Ω is holomorphically convex if

$$K \Subset \Omega \implies \widehat{K} \Subset \Omega.$$

Thus holomorphic functions prevent the hull of a compact set from escaping toward the

boundary.

[Example] Let $P \Subset U \subset \mathbb{C}^n$, where U is a neighbourhood of $\overline{P}$. Suppose that there are finitely many functions

$$f_1, \dots, f_N \in \mathcal{O}(U)$$

such that

$$P = \{z \in U : |f_1(z)| < 1, \dots, |f_N(z)| < 1\}.$$

Then P is called an analytic polyhedron.

Every analytic polyhedron is holomorphically convex. Indeed, let $K \Subset P$ and set

$$r_j := \sup_{z \in K} |f_j(z)| < 1, \quad 1 \leq j \leq N.$$

If $z \in \widehat{K}$, then the definition of the holomorphic hull, applied to each $f_j|_P \in \mathcal{O}(P)$, gives

$$|f_j(z)| \leq \sup_K |f_j| = r_j.$$

Therefore

$$\widehat{K} \subset \{z \in \overline{P} : |f_j(z)| \leq r_j \text{ for } 1 \leq j \leq N\} \Subset P.$$

The set on the right is compact because $\overline{P}$ is compact, and it lies inside P because every r_j is strictly smaller than 1. Thus $\widehat{K} \Subset P$.

[Lemma] If $\Omega \subset \mathbb{C}^n$ is holomorphically convex, then there is an exhaustion by compact sets

$$K_1 \Subset K_2 \Subset K_3 \Subset \dots, \quad \bigcup_{\nu=1}^{\infty} K_\nu = \Omega,$$

such that every K_ν is holomorphically convex in Ω :

$$\widehat{K_\nu} = K_\nu.$$

[Proof] Begin with an ordinary compact exhaustion

$$Q_1 \Subset Q_2 \Subset Q_3 \Subset \dots, \quad \bigcup_{\nu=1}^{\infty} Q_\nu = \Omega.$$

Set

$$K_1 := \widehat{Q_1}.$$

Holomorphic convexity says precisely that $K_1 \Subset \Omega$. Moreover, taking a holomorphic hull twice does not enlarge it:

$$\widehat{\widehat{Q_1}} = \widehat{Q_1}.$$

Indeed, for every $f \in \mathcal{O}(\Omega)$,

$$\sup_{\widehat{Q_1}} |f| = \sup_{Q_1} |f|.$$

Suppose K_ν has been constructed. Since K_ν is compact and the interiors of the Q_j exhaust Ω , choose m_ν large enough that

$$K_\nu \cup Q_{\nu+1} \Subset Q_{m_\nu}.$$

Define

$$K_{\nu+1} := \widehat{Q_{m_\nu}}.$$

Again, holomorphic convexity makes $K_{\nu+1}$ compactly contained in Ω , and idempotence of the hull gives

$$\widehat{\widehat{K_{\nu+1}}} = K_{\nu+1}.$$

Since $Q_{m_\nu} \subset K_{\nu+1}$, one also has

$$K_\nu \Subset K_{\nu+1}, \quad Q_{\nu+1} \Subset K_{\nu+1}.$$

Thus the K_ν have all the required properties.

[Lemma] A domain $\Omega \subset \mathbb{C}^n$ is holomorphically convex if and only if the following condition holds:

For every sequence $(z_\nu) \subset \Omega$ tending to $\partial\Omega$, there exists $f \in \mathcal{O}(\Omega)$ such that the sequence $(|f(z_\nu)|)$ is unbounded.

Here $z_\nu \rightarrow \partial\Omega$ means that (z_ν) eventually leaves every compact subset of Ω . When Ω is bounded, this is equivalent to

$$\text{dist}(z_\nu, \partial\Omega) \rightarrow 0.$$

[Proof] First assume that Ω is holomorphically convex, and choose the compact exhaustion (K_ν) from the preceding lemma. Given a sequence tending to $\partial\Omega$, pass to a subsequence and relabel the subsequence and the exhaustion so that

$$z_\nu \notin K_\nu, \quad z_\mu \in K_\nu \quad \text{whenever } \mu < \nu.$$

Because

$$K_\nu = \widehat{K_\nu}$$

and $z_\nu \notin K_\nu$, the definition of the hull gives a function $g_\nu \in \mathcal{O}(\Omega)$ satisfying

$$|g_\nu(z_\nu)| > \sup_{K_\nu} |g_\nu|.$$

After dividing by a suitable positive constant, obtain $h_\nu \in \mathcal{O}(\Omega)$ such that

$$\sup_{K_\nu} |h_\nu| < 1, \quad |h_\nu(z_\nu)| > 1.$$

Choose a sufficiently large positive integer m_ν and set

$$f_\nu := h_\nu^{m_\nu}$$

so that

$$\sup_{K_\nu} |f_\nu| < 2^{-\nu}$$

and, recursively,

$$|f_\nu(z_\nu)| > \nu + \sum_{\mu < \nu} |f_\mu(z_\nu)|.$$

The series

$$f := \sum_{\nu=1}^{\infty} f_\nu$$

converges uniformly on every compact subset of Ω , because on K_N every term with $\nu \geq N$ has norm at most $2^{-\nu}$. Hence $f \in \mathcal{O}(\Omega)$. Since $z_\nu \in K_\mu$ for every $\mu > \nu$,

$$\begin{aligned} |f(z_\nu)| &\geq |f_\nu(z_\nu)| - \sum_{\mu < \nu} |f_\mu(z_\nu)| - \sum_{\mu > \nu} |f_\mu(z_\nu)| \\ &> \nu - \sum_{\mu > \nu} 2^{-\mu} > \nu - 1. \end{aligned}$$

Thus $(|f(z_\nu)|)$ is unbounded on the selected subsequence and therefore on the original sequence.

Conversely, suppose the boundary-sequence condition holds but Ω is not holomorphically convex. Then there is a compact set $K \Subset \Omega$ for which

$$\widehat{K} \not\subset \Omega.$$

Choose a sequence

$$z_\nu \in \widehat{K}$$

that eventually leaves every compact subset of Ω . By assumption, some $f \in \mathcal{O}(\Omega)$ makes $(|f(z_\nu)|)$ unbounded. But the defining inequality for the hull gives

$$|f(z_\nu)| \leq \sup_K |f| < \infty$$

for every ν , a contradiction. Therefore Ω is holomorphically convex.

11 The Cartan-Thullen equivalence

[Theorem] For a domain $\Omega \subset \mathbb{C}^n$, the following are equivalent:

- 1. Ω is a domain of holomorphy.
- 2. Ω is holomorphically convex.
- 3. There exists one function $f \in \mathcal{O}(\Omega)$ that cannot be holomorphically extended across any point of $\partial\Omega$.
- 4. For every sequence $(z_\nu) \subset \Omega$ tending to $\partial\Omega$, there exists $f \in \mathcal{O}(\Omega)$ such that $(|f(z_\nu)|)$ is unbounded.

Here the third condition means that for every $p \in \partial\Omega$ there is no neighbourhood V of p and no $\tilde{f} \in \mathcal{O}(V)$ agreeing with f on a nonempty open subset of $V \cap \Omega$.

Implication 1 $\implies$ 2

[Proof] Use polydiscs for the boundary distance and set

$$\delta(z) = \sup \left\{ r > 0 : \overline{\Delta(z; r, \dots, r)} \subset \Omega \right\}.$$

Let $K \Subset \Omega$ and choose

$$0 < r < \inf_{z \in K} \delta(z).$$

The compact r -neighbourhood K_r formed from the corresponding closed polydiscs is contained in Ω . If $a \in \widehat{K}$, then every derivative $\partial^\alpha f$ is again holomorphic, so the definition of the hull followed by the Cauchy estimates gives

$$\begin{aligned} |\partial^\alpha f(a)| &\leq \sup_K |\partial^\alpha f| \\ &\leq \frac{\alpha!}{r^{|\alpha|}} \sup_{K_r} |f|. \end{aligned}$$

Therefore the Taylor series of every $f \in \mathcal{O}(\Omega)$ at a converges throughout $\Delta(a; r, \dots, r)$. If this polydisc crossed the boundary, these Taylor series would give a simultaneous continuation of all functions in $\mathcal{O}(\Omega)$ beyond Ω , contradicting the domain-of-holomorphy condition. Hence

$$\delta(a) \geq r \quad \text{for every } a \in \widehat{K}.$$

The coordinate functions also belong to $\mathcal{O}(\Omega)$, so the definition of the hull bounds every coordinate of a by its maximum on K . Thus $\widehat{K}$ is bounded and stays a positive distance from $\partial\Omega$. Since it is closed in Ω , it is compact in Ω .

Implication 2 $\implies$ 3

[Proof] Assume statement 2. By the holomorphically convex exhaustion lemma, choose compact sets

$$K_1 \Subset K_2 \Subset K_3 \Subset \cdots, \quad \bigcup_{\nu=1}^{\infty} K_\nu = \Omega, \quad \widehat{K_\nu} = K_\nu.$$

We first choose a sequence $(z_\nu) \subset \Omega$ such that

- $z_\nu \in \Omega \setminus K_\nu$ for every ν ;
- every point of $\partial\Omega$ is an accumulation point of (z_ν) , equivalently,

$$\partial\Omega \subset \overline{\{z_\nu : \nu \geq 1\}}.$$

To construct it, choose a sequence $(q_\nu) \subset \partial\Omega$ such that every boundary point is an accumulation point of (q_ν) . Since $K_\nu \Subset \Omega$, choose

$$z_\nu \in \Omega \setminus K_\nu, \quad |z_\nu - q_\nu| < \frac{1}{\nu}.$$

Thus every $a \in \partial\Omega$ is the limit of a subsequence of (z_ν) .

Since

$$z_\nu \notin K_\nu = \widehat{K_\nu},$$

the definition of the holomorphic hull gives $g_\nu \in \mathcal{O}(\Omega)$ such that

$$|g_\nu(z_\nu)| > \sup_{K_\nu} |g_\nu|.$$

Normalize and take a sufficiently large positive power. After relabelling the resulting function as f_ν , we may arrange that

$$f_\nu(z_\nu) = 1, \quad \sup_{K_\nu} |f_\nu| < 2^{-\nu}.$$

Indeed, first divide g_ν by $g_\nu(z_\nu)$ and then raise the quotient to a sufficiently large integer power.

Define

$$f(z) := \prod_{\nu=1}^{\infty} (1 - f_\nu(z))^\nu.$$

Fix N . For every $\nu \geq N$, the inclusion $K_N \subset K_\nu$ gives

$$\sup_{K_N} |f_\nu| < 2^{-\nu}.$$

Since

$$\sum_{\nu=N}^{\infty} \nu 2^{-\nu} < \infty,$$

the infinite product converges uniformly on K_N . Hence it converges uniformly on every compact subset of Ω and defines a holomorphic function

$$f \in \mathcal{O}(\Omega).$$

The function f is not identically zero. Indeed, choose an open ball $B \Subset K_N$. The finitely many functions $1 - f_\nu$ with $\nu < N$ are not identically zero, while

$$|f_\nu| < 1 \quad \text{on } K_N \quad \text{for } \nu \geq N.$$

One may therefore choose $z_0 \in B$ outside the zero sets of the finitely many initial factors, and the convergent tail product is nonzero at z_0 . Thus $f(z_0) \neq 0$.

At z_ν , the factor

$$(1 - f_\nu)^\nu$$

vanishes to order at least ν . Consequently,

$$\partial^\alpha f(z_\nu) = 0 \quad \text{whenever } |\alpha| < \nu.$$

Suppose that f extends holomorphically across some $z^* \in \partial\Omega$. Since every boundary point is an accumulation point of (z_ν) , choose a subsequence

$$z_{\nu_j} \longrightarrow z^*.$$

For every fixed multi-index α , one has $|\alpha| < \nu_j$ for all sufficiently large j , and therefore

$$\partial^\alpha f(z_{\nu_j}) = 0.$$

Continuity of the derivatives of the extension gives

$$\partial^\alpha f(z^*) = 0 \quad \text{for every multi-index } \alpha.$$

The Taylor series of the extension at z^* is therefore identically zero, so the extension vanishes on a neighbourhood of z^* . The identity theorem then forces $f \equiv 0$ on Ω , contradicting the construction above. Thus f cannot be extended through any boundary point, proving $2 \implies 3$.

Implications $3 \implies 1$ and $2 \iff 4$

[Proof] For $3 \implies 1$, let $\Omega' \supsetneq \Omega$ be a strictly larger domain. A path in Ω' from a point of Ω to a point of $\Omega' \setminus \Omega$ meets some

$$a \in \partial\Omega \cap \Omega'.$$

If the restriction map $\mathcal{O}(\Omega') \rightarrow \mathcal{O}(\Omega)$ were surjective, the function in statement 3 would extend to Ω' and hence across a , a contradiction. The boundary-sequence lemma gives $2 \iff 4$. Therefore

$$1 \implies 2 \implies 3 \implies 1, \quad 2 \iff 4,$$

so all four statements are equivalent.

Day 3

12 Smooth boundaries, the Levi form, and pseudoconvexity

Let $\Omega \subset \mathbb{C}^n$ be a domain with smooth boundary.

[Definition] A smooth real-valued function

$$\rho \in C^\infty(U, \mathbb{R})$$

on a neighbourhood U of $p \in \partial\Omega$ is called a local defining function for Ω near p if

$$\Omega \cap U = \{z \in U : \rho(z) < 0\}, \quad \partial\Omega \cap U = \{z \in U : \rho(z) = 0\},$$

and

$$d\rho \neq 0 \quad \text{on } \partial\Omega \cap U.$$

Because ρ is real-valued, the last condition is equivalent to

$$\partial\rho = \sum_{j=1}^n \frac{\partial\rho}{\partial z_j} dz_j \neq 0 \quad \text{on } \partial\Omega \cap U.$$

The sign convention $\rho < 0$ inside Ω fixes the outward direction of the defining function.

[Definition] Let $p \in \partial\Omega$, and let ρ be a local defining function near p . The complex tangent space of $\partial\Omega$ at p is

$$T_p^{1,0}(\partial\Omega) = \left\{ \xi = (\xi_1, \dots, \xi_n) \in \mathbb{C}^n : \sum_{j=1}^n \frac{\partial\rho}{\partial z_j}(p) \xi_j = 0 \right\}.$$

Thus a complex direction ξ is tangent precisely when the first-order variation of ρ in that direction vanishes.

[Definition] The Levi form of ρ at p is the Hermitian form

$$\mathcal{L}_\rho(p; \xi) := \sum_{j,k=1}^n \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k}(p) \xi_j \bar{\xi}_k.$$

Equivalently, if

$$H_{\mathbb{C}}(\rho)(p) = \left(\frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k}(p) \right)_{j,k},$$

then

$$\mathcal{L}_\rho(p; \xi) = \xi^* H_{\mathbb{C}}(\rho)(p) \xi.$$

This matrix is the complex Hessian of ρ .

[Definition] The boundary $\partial\Omega$ is pseudoconvex at p if

$$\mathcal{L}_\rho(p; \xi) \geq 0 \quad \text{for every } \xi \in T_p^{1,0}(\partial\Omega).$$

The domain Ω is pseudoconvex if this condition holds at every point of $\partial\Omega$.

[Lemma] Pseudoconvexity is independent of the choice of local defining function.

[Proof] Let ρ and $\tilde{\rho}$ be two local defining functions having the same sign inside Ω . After shrinking the neighbourhood if necessary, there is a smooth positive function h such that

$$\tilde{\rho} = h\rho.$$

At a boundary point p , since $\rho(p) = 0$,

$$\partial\tilde{\rho}(p) = h(p)\partial\rho(p).$$

Hence ρ and $\tilde{\rho}$ determine the same complex tangent space. Differentiating once more gives

$$\frac{\partial^2 \tilde{\rho}}{\partial z_j \partial \bar{z}_k} = h \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k} + \frac{\partial h}{\partial z_j} \frac{\partial \rho}{\partial \bar{z}_k} + \frac{\partial \rho}{\partial z_j} \frac{\partial h}{\partial \bar{z}_k} + \rho \frac{\partial^2 h}{\partial z_j \partial \bar{z}_k}.$$

If $\xi \in T_p^{1,0}(\partial\Omega)$, the middle two terms vanish after contraction with $\xi_j \bar{\xi}_k$, and the last term vanishes because $\rho(p) = 0$. Therefore

$$\mathcal{L}_{\tilde{\rho}}(p; \xi) = h(p) \mathcal{L}_\rho(p; \xi).$$

Since $h(p) > 0$, the two Levi forms have the same sign on the complex tangent space. Thus pseudoconvexity is a geometric property of the boundary.

[Theorem] Let $\Omega \subset \mathbb{C}^n$ be a domain with smooth boundary. Then

$$\Omega \text{ is a domain of holomorphy} \iff \Omega \text{ is pseudoconvex.}$$

The difficult implication from pseudoconvexity to the existence of sufficiently many holomorphic functions is known as the Levi problem.

13 Complex manifolds and complex subspaces

[Definition] A complex manifold of complex dimension n is a real smooth manifold X of real dimension $2n$ together with an atlas of charts

$$\varphi_\alpha : U_\alpha \longrightarrow V_\alpha \subset \mathbb{C}^n$$

such that every transition map

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \longrightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

is biholomorphic wherever it is defined. Thus the different coordinate descriptions are compatible with the complex structure of $\mathbb{C}^n$.

[Definition] Let X and Y be complex manifolds. A map $F : X \rightarrow Y$ is holomorphic if for every pair of charts $\varphi : U \rightarrow \mathbb{C}^n$ on X and $\psi : V \rightarrow \mathbb{C}^m$ on Y , the coordinate representation

$$\psi \circ F \circ \varphi^{-1}$$

is holomorphic on its domain. In particular, a function $f : X \rightarrow \mathbb{C}$ is holomorphic exactly when $f \circ \varphi^{-1}$ is holomorphic in every local chart.

[Definition] A bijection $F : X \rightarrow Y$ is a biholomorphism if both F and F^{-1} are holomorphic. In that case X and Y are biholomorphic, written

$$X \cong_{\text{hol}} Y.$$

[Example] Let

$$\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2 \subset \mathbb{C},$$

where ω_1 and ω_2 are linearly independent over $\mathbb{R}$. Translation by elements of Λ is holomorphic, so the quotient

$$\mathbb{C}/\Lambda$$

inherits the structure of a compact complex manifold of complex dimension one. It is a one-dimensional complex torus.

[Proposition] Let $U \subset \mathbb{C}^n$ be open and let $f \in \mathcal{O}(U)$. Suppose

$$df_p \neq 0 \quad \text{for every } p \in f^{-1}(0).$$

Then the zero set

$$Z(f) = \{z \in U : f(z) = 0\}$$

is a complex submanifold of U of complex dimension $n - 1$.

[Proof] Fix $p \in Z(f)$. After relabelling the coordinates, assume

$$\frac{\partial f}{\partial z_n}(p) \neq 0.$$

The holomorphic implicit function theorem gives neighbourhoods of p and $(p_1, \dots, p_{n-1})$, together with a holomorphic function g , such that locally

$$Z(f) = \{(z_1, \dots, z_{n-1}, z_n) : z_n = g(z_1, \dots, z_{n-1})\}.$$

Projection onto the first $n - 1$ coordinates is therefore a holomorphic chart on $Z(f)$. Hence $Z(f)$ is a complex hypersurface.

[Remark] More generally, if $F : U \rightarrow \mathbb{C}^k$ is holomorphic and has complex rank k at every point of $F^{-1}(0)$, then $F^{-1}(0)$ is a complex submanifold of codimension k .

[Example] Complex projective space is

$$\mathbb{P}^n = (\mathbb{C}^{n+1} \setminus \{0\}) / \sim,$$

where

$$(z_0, \dots, z_n) \sim (w_0, \dots, w_n)$$

if and only if there is some $\lambda \in \mathbb{C}^*$ such that

$$(z_0, \dots, z_n) = \lambda(w_0, \dots, w_n).$$

Write an equivalence class as $[z_0 : \dots : z_n]$. For $0 \leq i \leq n$, set

$$U_i = \{[z_0 : \dots : z_n] \in \mathbb{P}^n : z_i \neq 0\}.$$

The map

$$\varphi_i : U_i \longrightarrow \mathbb{C}^n, \quad [z_0 : \dots : z_n] \longmapsto \left(\frac{z_0}{z_i}, \dots, \widehat{\frac{z_i}{z_i}}, \dots, \frac{z_n}{z_i} \right)$$

is a chart; the hat means that the i -th entry is omitted. The transition maps consist of ratios of coordinates with nonzero denominators and are biholomorphic. These $n + 1$ charts give $\mathbb{P}^n$ the structure of a compact complex manifold of dimension n .

[Definition] Let X be a complex manifold of dimension n . A subset $Y \subset X$ is a complex analytic submanifold of dimension m if, for every $p \in Y$, there is a holomorphic chart $\varphi : U \rightarrow V \subset \mathbb{C}^n$ such that

$$\varphi(Y \cap U) = V \cap (\mathbb{C}^m \times \{0\}).$$

Thus suitable holomorphic coordinates identify Y locally with the coordinate plane

$$z_{m+1} = \dots = z_n = 0.$$

[Definition] A closed subset $A \subset X$ is a complex analytic subvariety if, for every $p \in X$, there is a neighbourhood U of p and finitely many holomorphic functions

$$f_1, \dots, f_k \in \mathcal{O}(U)$$

such that

$$A \cap U = \{z \in U : f_1(z) = \dots = f_k(z) = 0\}.$$

If A is locally defined by one nonzero holomorphic equation, it is called a complex analytic hypersurface.

[Remark] A complex submanifold has no singular points, because it is locally a coordinate plane. A complex analytic subvariety may have singular points when the defining equations fail the full-rank condition. For example,

$$A = \{(z, w) \in \mathbb{C}^2 : w^2 = z^3\}$$

is a complex analytic subvariety, but the origin is singular because the differential of $w^2 - z^3$ vanishes there.

[Lemma] There is no compact, connected, positive-dimensional complex submanifold of $\mathbb{C}^N$.

[Proof] Suppose that $M \subset \mathbb{C}^N$ were such a submanifold. Each ambient coordinate function restricts to a holomorphic function

$$z_j|_M : M \longrightarrow \mathbb{C}.$$

Because M is compact, $|z_j|$ attains a maximum. The maximum modulus principle, applied in a local chart and then propagated through the connected manifold, implies that every $z_j|_M$ is constant. Hence the inclusion

$$M \hookrightarrow \mathbb{C}^N$$

is constant, contradicting the assumption that M has positive dimension.

[Remark] The positive-dimensional qualification is necessary: a single point is a compact zero-dimensional complex submanifold of $\mathbb{C}^N$.

14 Almost-complex structures and integrability

Two different existence questions should be separated:

- whether a smooth manifold admits an almost-complex structure;
- whether an almost-complex structure comes from genuine holomorphic coordinates.

The first is primarily topological, whereas the second is an additional local differential condition.

[Definition] Let X be a smooth manifold. An almost-complex structure on X is a smooth bundle endomorphism

$$J : TX \longrightarrow TX$$

such that

$$J^2 = -\text{Id}_{TX}.$$

Equivalently, every fiber map

$$J_p : T_p X \longrightarrow T_p X$$

is real-linear and satisfies $J_p^2 = -\text{Id}$. In particular, X must have even real dimension. The existence of J is equivalent to reducing the structure group of TX from $GL(2n, \mathbb{R})$ to $GL(n, \mathbb{C})$.

[Remark] The existence of an almost-complex structure does not imply the existence of a complex structure. The sphere S^6 carries a natural almost-complex structure, but whether S^6 admits any integrable complex structure remains an open problem.

[Example] If X is already a complex manifold and

$$z_j = x_j + iy_j$$

are local holomorphic coordinates, define

$$J \left(\frac{\partial}{\partial x_j} \right) = \frac{\partial}{\partial y_j}, \quad J \left(\frac{\partial}{\partial y_j} \right) = -\frac{\partial}{\partial x_j}.$$

The differentials of holomorphic transition maps satisfy the Cauchy–Riemann equations and therefore commute with this standard action of J . Hence the locally defined endomorphisms agree on chart overlaps and determine a global almost-complex structure.

[Definition] An almost-complex structure J on X is integrable if there is a complex-manifold atlas on X whose induced almost-complex structure is J . Thus integrability asks whether one can find local coordinates in which J becomes the standard multiplication-by- i structure.

Complexification and the type decomposition

[Definition] The complexified tangent bundle is

$$T_{\mathbb{C}}X := TX \otimes_{\mathbb{R}} \mathbb{C}.$$

Extend J complex-linearly to $T_{\mathbb{C}}X$. Since $J^2 = -\text{Id}$, its only eigenvalues are i and $-i$. Define

$$T^{1,0}X := \ker(J - i\text{Id}), \quad T^{0,1}X := \ker(J + i\text{Id}).$$

Then

$$T_{\mathbb{C}}X = T^{1,0}X \oplus T^{0,1}X.$$

The corresponding projections are

$$\pi^{1,0} = \frac{1}{2} (\text{Id} - iJ), \quad \pi^{0,1} = \frac{1}{2} (\text{Id} + iJ).$$

Complex conjugation interchanges $T^{1,0}X$ and $T^{0,1}X$, so both have complex rank n when $\dim_{\mathbb{R}} X = 2n$.

[Example] For the standard complex structure in local coordinates $z_j = x_j + iy_j$,

$$\frac{\partial}{\partial z_j} = \frac{1}{2} \left(\frac{\partial}{\partial x_j} - i \frac{\partial}{\partial y_j} \right), \quad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2} \left(\frac{\partial}{\partial x_j} + i \frac{\partial}{\partial y_j} \right),$$

and

$$T^{1,0}X = \text{span}_{\mathbb{C}} \left\{ \frac{\partial}{\partial z_1}, \dots, \frac{\partial}{\partial z_n} \right\},$$

$$T^{0,1}X = \text{span}_{\mathbb{C}} \left\{ \frac{\partial}{\partial \bar{z}_1}, \dots, \frac{\partial}{\partial \bar{z}_n} \right\}.$$

[Theorem] An almost-complex structure J is integrable if and only if the space of $(1,0)$ vector fields is closed under the Lie bracket:

$$[\Gamma(T^{1,0}X), \Gamma(T^{1,0}X)] \subset \Gamma(T^{1,0}X).$$

This condition is called involutivity of $T^{1,0}X$.

[Proof sketch] If J comes from complex coordinates, then $T^{1,0}X$ is locally spanned by the commuting coordinate fields $\partial/\partial z_j$. The Leibniz rule for the Lie bracket then proves involutivity.

The converse is the difficult direction. Involutivity is exactly the compatibility condition needed to construct locally n independent functions

$$f_1, \dots, f_n$$

satisfying

$$\bar{\partial}_J f_j = 0$$

and

$$df_1 \wedge \dots \wedge df_n \neq 0.$$

The map $(f_1, \dots, f_n)$ then supplies a holomorphic coordinate chart in which J is standard.

Forms of type (p, q) and the operator $\bar{\partial}_J$

The eigenspace decomposition of $T_{\mathbb{C}}X$ induces a decomposition of complex differential forms. Define

$$\Lambda^{p,q}T^*X := \Lambda^p(T^{1,0}X)^* \wedge \Lambda^q(T^{0,1}X)^*.$$

Then

$$\Lambda^k T_{\mathbb{C}}^*X = \bigoplus_{p+q=k} \Lambda^{p,q}T^*X.$$

Write $\Omega_J^{p,q}(X)$ for the smooth sections of $\Lambda^{p,q}T^*X$, and let

$$\pi^{p,q} : \Omega^k(X; \mathbb{C}) \longrightarrow \Omega_J^{p,q}(X)$$

denote projection onto the (p, q) component.

[Definition] For $\alpha \in \Omega_J^{p,q}(X)$, define

$$\partial_J \alpha := \pi^{p+1,q}(d\alpha), \quad \bar{\partial}_J \alpha := \pi^{p,q+1}(d\alpha).$$

If J is induced by a complex-manifold structure, these are the ordinary Dolbeault operators ∂ and $\bar{\partial}$.

[Proposition] The following conditions are equivalent:

1. J is integrable.
2. $T^{1,0}X$ is involutive.
3. $\bar{\partial}_J^2 f = 0$ for every $f \in C^\infty(X; \mathbb{C})$.

When these conditions hold,

$$d = \partial_J + \bar{\partial}_J$$

and

$$\partial_J^2 = 0, \quad \bar{\partial}_J^2 = 0, \quad \partial_J \bar{\partial}_J + \bar{\partial}_J \partial_J = 0$$

on all complex differential forms.

[Remark] For a general almost-complex structure, the formula $\bar{\partial}_J = \pi^{p,q+1} \circ d$ does not imply $\bar{\partial}_J^2 = 0$: the projection does not commute with d . Nonintegrability appears through additional components of d of types $(p+2, q-1)$ and $(p-1, q+2)$. Their vanishing is equivalent to the involutivity condition above.

Day 4

15 Holomorphic vector bundles

The complexified tangent bundle introduced on Day 3 was

$$T_{\mathbb{C}}X = TX \otimes_{\mathbb{R}} \mathbb{C} = T^{1,0}X \oplus T^{0,1}X.$$

It has complex rank $2n$ when $\dim_{\mathbb{C}} X = n$. By contrast, the **holomorphic tangent bundle** of a complex manifold is the rank- n bundle

$$T^{1,0}X = \ker(J - i \text{Id}) \subset T_{\mathbb{C}}X.$$

Thus the full complexification and the holomorphic tangent bundle are not the same bundle: the latter is one of the two eigensubbundles of the former. There is, however, a canonical complex-linear isomorphism

$$(TX, J) \longrightarrow T^{1,0}X, \quad v \longmapsto v^{1,0} := \frac{1}{2}(v - iJv),$$

where (TX, J) means that each real tangent space is regarded as a complex vector space by letting i act as J . Indeed,

$$Jv^{1,0} = \frac{1}{2}(Jv + iv) = iv^{1,0},$$

so the image lies in $T^{1,0}X$. Its inverse sends $Z \in T^{1,0}X$ to the real vector $Z + \overline{Z}$, since

$$(Z + \overline{Z})^{1,0} = Z.$$

Thus it is an isomorphism, and both sides have complex dimension n . This explains why the holomorphic tangent space is sometimes written simply as T_pX after the complex structure has been understood, even though its realization inside the complexification is $T_p^{1,0}X$.

[Example] In holomorphic coordinates $z = (z_1, \dots, z_n)$, the bundle $T^{1,0}X$ has the local frame

$$\frac{\partial}{\partial z_1}, \dots, \frac{\partial}{\partial z_n}.$$

This is exactly the $+i$ eigenspace description from Day 3, because

$$J\left(\frac{\partial}{\partial z_j}\right) = i\frac{\partial}{\partial z_j}.$$

If $w = (w_1, \dots, w_n)$ is another holomorphic coordinate system, then

$$\frac{\partial}{\partial z_i} = \sum_{j=1}^n \frac{\partial w_j}{\partial z_i} \frac{\partial}{\partial w_j}.$$

The coefficient matrix is the holomorphic Jacobian of the coordinate change. Hence these local frames give $T^{1,0}X$ the structure of a holomorphic vector bundle. The coordinate-frame construction on Day 4 and the eigensubbundle construction on Day 3 therefore define the same holomorphic tangent bundle.

[Example] The holomorphic cotangent bundle is

$$T^{*1,0}X := (T^{1,0}X)^*.$$

In holomorphic coordinates it has the local frame

$$dz_1, \dots, dz_n.$$

If $w = (w_1, \dots, w_n)$ is another holomorphic coordinate system, then

$$dz_i = \sum_{j=1}^n \frac{\partial z_i}{\partial w_j} dw_j.$$

Thus the cotangent transition matrix is the inverse transpose of the tangent transition matrix, as expected for a dual bundle.

[Definition] Let X be a complex manifold. A holomorphic vector bundle of rank r over X consists of a complex manifold E and a holomorphic surjection

$$\pi : E \longrightarrow X$$

such that every fiber

$$E_p := \pi^{-1}(p)$$

is a complex vector space of dimension r , and every $p \in X$ has an open neighbourhood U with a biholomorphism

$$\Phi : \pi^{-1}(U) \longrightarrow U \times \mathbb{C}^r$$

satisfying

$$\text{pr}_1 \circ \Phi = \pi.$$

Moreover, for each $q \in U$, the restriction

$$\Phi|_{E_q} : E_q \longrightarrow \{q\} \times \mathbb{C}^r$$

must be a complex-linear isomorphism. The map Φ is called a holomorphic local trivialization.

[Definition] A holomorphic section of E over an open set $U \subset X$ is a holomorphic map

$$s : U \longrightarrow E$$

such that

$$\pi \circ s = \text{Id}_U.$$

The space of such sections is denoted by

$$\Gamma_{\text{hol}}(U, E).$$

[Remark] A local holomorphic section does not necessarily extend to a global holomorphic section. For example, for the trivial line bundle over $\mathbb{C}$, the section represented by

$$s(z) = \frac{1}{z}$$

on $\mathbb{C} \setminus \{0\}$ has no holomorphic extension across the origin.

[Definition] A holomorphic local frame of E over U is an ordered collection

$$(e_1, \dots, e_r), \quad e_a \in \Gamma_{\text{hol}}(U, E),$$

such that

$$e_1(p), \dots, e_r(p)$$

form a complex basis of E_p for every $p \in U$.

[Lemma] A holomorphic local trivialization of E over U is equivalent to a holomorphic local frame over U .

[Proof] Let $\varepsilon_1, \dots, \varepsilon_r$ be the standard basis of $\mathbb{C}^r$. Given a trivialization Φ , define

$$e_a(p) := \Phi^{-1}(p, \varepsilon_a).$$

These are holomorphic sections and form a basis in every fiber.

Conversely, given a holomorphic frame $(e_1, \dots, e_r)$, every $v \in E_p$ has a unique expression

$$v = \sum_{a=1}^r v^a e_a(p).$$

Define

$$\Phi(v) = (p, (v^1, \dots, v^r)).$$

This is a fiberwise complex-linear biholomorphism

$$\Phi : \pi^{-1}(U) \longrightarrow U \times \mathbb{C}^r.$$

[Definition] Choose holomorphic local trivializations $\Phi_i : \pi^{-1}(U_i) \longrightarrow U_i \times \mathbb{C}^r$. On an overlap $U_i \cap U_j$, the change of trivialization has the form $\Phi_i \circ \Phi_j^{-1}(p, v) = (p, \psi_{ij}(p)v)$, where $\psi_{ij} : U_i \cap U_j \longrightarrow GL(r, \mathbb{C})$ is holomorphic. The maps ψ_{ij} are called the transition functions of E .

[Lemma] The transition functions satisfy the cocycle identities

$$\psi_{ii} = \text{Id}, \quad \psi_{ij}\psi_{ji} = \text{Id},$$

and, on every triple overlap,

$$\psi_{ij}\psi_{jk} = \psi_{ik}.$$

Equivalently,

$$\psi_{ij}\psi_{jk}\psi_{ki} = \text{Id}.$$

[Proof] These identities follow by composing the corresponding changes of trivialization. For example,

$$\Phi_i \circ \Phi_j^{-1} \circ \Phi_j \circ \Phi_k^{-1} = \Phi_i \circ \Phi_k^{-1},$$

which gives $\psi_{ij}\psi_{jk} = \psi_{ik}$.

[Proposition] Conversely, let (U_i) be an open cover of X , and suppose that holomorphic maps

$$\psi_{ij} : U_i \cap U_j \longrightarrow GL(r, \mathbb{C})$$

satisfy the cocycle identities above. Then they determine a holomorphic vector bundle of rank r over X .

[Proof] Form the disjoint union

$$\bigsqcup_i U_i \times \mathbb{C}^r$$

and impose the equivalence relation

$$(p, v)_j \sim (p, \psi_{ij}(p)v)_i \quad \text{for } p \in U_i \cap U_j.$$

The cocycle identities make this an equivalence relation. The quotient

$$E := \left(\bigsqcup_i U_i \times \mathbb{C}^r \right) / \sim$$

has a well-defined projection to X . Each copy $U_i \times \mathbb{C}^r$ gives a holomorphic local trivialization, and the prescribed functions ψ_{ij} are exactly its transition functions.

[Remark] Relative to local frames e_i on U_i , a section s of a line bundle is represented by holomorphic functions $f_i \in \mathcal{O}(U_i)$ through

$$s|_{U_i} = f_i e_i.$$

If the change of trivialization from the j -th coordinates to the i -th coordinates is multiplication by ψ_{ij} , then the local functions satisfy

$$f_i = \psi_{ij} f_j$$

on $U_i \cap U_j$. Conversely, any family (f_i) satisfying these relations determines a global holomorphic section.

Standard constructions with holomorphic vector bundles

[Definition] If E and F are holomorphic vector bundles over X , then the following bundles inherit natural holomorphic structures from their transition functions:

$$E \oplus F, \quad E \otimes F, \quad E^*, \quad \operatorname{Sym}^m E, \quad \Lambda^m E, \quad \operatorname{Hom}(E, F).$$

Their fibers are the corresponding linear-algebraic constructions on E_p and F_p . For example,

$$\operatorname{Hom}(E, F) \cong E^* \otimes F.$$

[Definition] Let $f : X \rightarrow Y$ be holomorphic and let $E \rightarrow Y$ be a holomorphic vector bundle. The pullback bundle $f^*E \rightarrow X$ is

$$f^*E = \{(p, v) \in X \times E : f(p) = \pi(v)\},$$

with fiber

$$(f^*E)_p = E_{f(p)}.$$

Pulling back holomorphic local trivializations of E gives f^*E its holomorphic bundle structure.

The bundles $\mathcal{O}_{\mathbb{P}^n}(m)$

[Example] The tautological line bundle on projective space is

$$\mathcal{O}_{\mathbb{P}^n}(-1) = \left\{ ([z], v) \in \mathbb{P}^n \times \mathbb{C}^{n+1} : v \in \mathbb{C}z \right\}.$$

Its fiber over $[z]$ is the line represented by $[z]$:

$$\mathcal{O}_{\mathbb{P}^n}(-1)_{[z]} = \mathbb{C}z \cong \mathbb{C}.$$

On the standard affine chart

$$U_i = \{[z_0 : \cdots : z_n] : z_i \neq 0\},$$

define the local frame

$$e_i([z]) := \frac{z}{z_i}.$$

On $U_i \cap U_j$ one has

$$e_j = \frac{z_i}{z_j} e_i.$$

Therefore the transition function from the j -th fiber coordinate to the i -th fiber coordinate is

$$\psi_{ij} = \frac{z_i}{z_j}.$$

The ratios are unchanged when z is rescaled, so they are well-defined holomorphic functions on projective space.

[Definition] The hyperplane bundle is the dual tautological bundle

$$\mathcal{O}_{\mathbb{P}^n}(1) := \mathcal{O}_{\mathbb{P}^n}(-1)^*.$$

Its transition functions are

$$\psi_{ij}^{(1)} = \frac{z_j}{z_i}.$$

For $m \in \mathbb{Z}$, define

$$\mathcal{O}_{\mathbb{P}^n}(m) = \begin{cases} \mathcal{O}_{\mathbb{P}^n}(1)^{\otimes m}, & m > 0, \\ \mathcal{O}_{\mathbb{P}^n}, & m = 0, \\ \mathcal{O}_{\mathbb{P}^n}(-1)^{\otimes (-m)}, & m < 0. \end{cases}$$

Its transition functions are

$$\psi_{ij}^{(m)} = \left(\frac{z_j}{z_i} \right)^m.$$

[Theorem] Assume $n \geq 1$. Then

$$\dim H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m)) = \begin{cases} 0, & m < 0, \\ \binom{n+m}{m}, & m \geq 0. \end{cases}$$

For $m \geq 0$, the global sections are naturally identified with homogeneous polynomials of degree m in $z_0, \dots, z_n$.

[Proof] Let s be a global section and write

$$s|_{U_i} = f_i e_i^{(m)}$$

in the standard local frame of $\mathcal{O}(m)$. On overlaps, the local functions obey

$$\frac{f_i}{f_j} = \left(\frac{z_j}{z_i} \right)^m,$$

or equivalently,

$$f_i z_i^m = f_j z_j^m.$$

Thus the expressions

$$F(z) := f_i([z]) z_i^m$$

agree and define a holomorphic function on $\mathbb{C}^{n+1} \setminus \{0\}$ satisfying

$$F(\lambda z) = \lambda^m F(z)$$

for every $\lambda \in \mathbb{C}^*$. Hartogs extension gives a holomorphic extension across the origin. If $m \geq 0$, the homogeneity forces this extension to be a homogeneous polynomial of degree m . Conversely, every such polynomial F gives a section by

$$f_i([z]) = \frac{F(z)}{z_i^m}$$

on U_i . The degree- m monomials form a basis, and their number is

$$\binom{n+m}{m}.$$

If $m < 0$, the extended function is holomorphic at the origin and homogeneous of negative degree. For fixed $z \neq 0$, the identity

$$F(\lambda z) = \lambda^m F(z)$$

as $\lambda \rightarrow 0$ forces $F(z) = 0$. Hence there are no nonzero global sections.

16 de Rham and Dolbeault cohomology

[Definition] A cochain complex is a sequence of abelian groups or vector spaces

$$\dots \longrightarrow C^{k-1} \xrightarrow{d^{k-1}} C^k \xrightarrow{d^k} C^{k+1} \longrightarrow \dots$$

such that

$$d^k \circ d^{k-1} = 0.$$

Its k -th cohomology group is

$$H^k(C^\bullet, d) := \frac{\ker d^k}{\operatorname{im} d^{k-1}}.$$

Thus cohomology measures the failure of every closed element to be exact.

[Definition] Let X be a smooth manifold. Adopt the convention

$$\mathcal{A}^p(X) = \begin{cases} 0, & p < 0, \\ \text{complex-valued smooth } p\text{-forms on } X, & p \geq 0. \end{cases}$$

The de Rham complex is

$$0 \longrightarrow \mathcal{A}^0(X) \xrightarrow{d} \mathcal{A}^1(X) \xrightarrow{d} \mathcal{A}^2(X) \xrightarrow{d} \dots,$$

and its cohomology is

$$H_{\text{dR}}^p(X; \mathbb{C}) = \frac{\ker(d : \mathcal{A}^p(X) \rightarrow \mathcal{A}^{p+1}(X))}{\operatorname{im}(d : \mathcal{A}^{p-1}(X) \rightarrow \mathcal{A}^p(X))}.$$

Using real-valued forms instead gives $H_{\text{dR}}^p(X; \mathbb{R})$.

[Definition] If X is a complex manifold, set

$$\mathcal{A}^{p,q}(X) = \begin{cases} 0, & p < 0 \text{ or } q < 0, \\ \text{complex-valued smooth } (p, q)\text{-forms on } X, & p, q \geq 0. \end{cases}$$

For fixed p , the Dolbeault complex is

$$0 \longrightarrow \mathcal{A}^{p,0}(X) \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1}(X) \xrightarrow{\bar{\partial}} \mathcal{A}^{p,2}(X) \xrightarrow{\bar{\partial}} \dots$$

Its cohomology groups are

$$H_{\bar{\partial}}^{p,q}(X) = \frac{\ker(\bar{\partial} : \mathcal{A}^{p,q}(X) \rightarrow \mathcal{A}^{p,q+1}(X))}{\operatorname{im}(\bar{\partial} : \mathcal{A}^{p,q-1}(X) \rightarrow \mathcal{A}^{p,q}(X))}.$$

[Theorem] Poincare and Dolbeault lemmas. If $B \subset \mathbb{C}^n$ is a ball, then

$$H_{\text{dR}}^p(B; \mathbb{C}) = 0 \quad \text{for } p \geq 1,$$

and

$$H_{\bar{\partial}}^{p,q}(B) = 0 \quad \text{for } q \geq 1.$$

Equivalently, every d -closed positive-degree form on a ball is locally d -exact, and every $\bar{\partial}$ -closed (p, q) -form with $q > 0$ is locally $\bar{\partial}$ -exact.

[Remark] Sheaf cohomology is motivated by local-to-global problems such as the Mittag-Leffler problem. Let X be a compact Riemann surface, choose points $p_1, \dots, p_N \in X$, and prescribe at each p_i a finite principal part

$$P_i(t_i) = \sum_{k=1}^{m_i} \frac{c_k^{(i)}}{t_i^k}$$

in a local coordinate t_i centered at p_i . The problem is to find a global meromorphic function F on X whose Laurent expansion at each p_i has the prescribed principal part.

Choose an open cover (U_i) and local meromorphic functions f_i carrying the desired principal parts. On overlaps their differences are holomorphic:

$$f_{ij} := f_i - f_j \in \mathcal{O}(U_i \cap U_j).$$

To construct one global meromorphic function with these principal parts, one seeks holomorphic functions $h_i \in \mathcal{O}(U_i)$ satisfying

$$h_i - h_j = f_{ij}.$$

The overlap functions obey the cocycle identity

$$f_{ij} + f_{ji} = 0,$$

and, on triple overlaps,

$$f_{ij} + f_{jk} + f_{ki} = 0,$$

and the problem is whether this cocycle is a coboundary. If suitable h_i exist, then the functions

$$F_i := f_i - h_i$$

agree on overlaps and glue to the desired global meromorphic function. Cohomology records precisely the obstruction to solving such gluing problems.

17 Presheaves and sheaves

[Definition] A presheaf $\mathcal{F}$ of abelian groups on a topological space X assigns an abelian group $\mathcal{F}(U)$ to every open set $U \subset X$ and, whenever $V \subset U$, a restriction homomorphism

$$\rho_V^U : \mathcal{F}(U) \longrightarrow \mathcal{F}(V).$$

These maps must satisfy

$$\rho_U^U = \text{Id}_{\mathcal{F}(U)}, \quad \rho_W^V \circ \rho_V^U = \rho_W^U$$

whenever $W \subset V \subset U$. An element of $\mathcal{F}(U)$ is called a section of $\mathcal{F}$ over U .

[Definition] The stalk of a presheaf $\mathcal{F}$ at $p \in X$ is the direct limit

$$\mathcal{F}_p := \varinjlim_{p \in U} \mathcal{F}(U).$$

Concretely, it is the set of equivalence classes of pairs (U, s) with $p \in U$ and $s \in \mathcal{F}(U)$, where

$$(U, s) \sim (V, t)$$

if there is an open neighbourhood $W \subset U \cap V$ of p such that

$$s|_W = t|_W.$$

An element of $\mathcal{F}_p$ is called a germ of a section at p .

[Definition] A morphism of presheaves

$$\Phi : \mathcal{F} \longrightarrow \mathcal{G}$$

is a family of homomorphisms

$$\Phi_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

that commutes with restrictions: whenever $V \subset U$,

$$\rho_{V, \mathcal{G}}^U \circ \Phi_U = \Phi_V \circ \rho_{V, \mathcal{F}}^U.$$

A morphism of sheaves is a morphism of their underlying presheaves.

[Definition] A presheaf $\mathcal{F}$ is a sheaf if it satisfies the following local-to-global property. Let

$$U = \bigcup_{i \in I} U_i$$

be an open cover and suppose that sections $s_i \in \mathcal{F}(U_i)$ agree on every overlap:

$$\rho_{U_i \cap U_j}^{U_i}(s_i) = \rho_{U_i \cap U_j}^{U_j}(s_j).$$

Then there exists a unique section $s \in \mathcal{F}(U)$ such that

$$\rho_{U_i}^U(s) = s_i$$

for every i . Existence is the gluing axiom and uniqueness is the locality axiom.

[Example] The assignments

$$U \mapsto C^\infty(U), \quad U \mapsto \mathcal{O}(U), \quad U \mapsto \Omega^k(U)$$

with the ordinary restriction maps are sheaves. Smooth functions, holomorphic functions, and differential forms can all be checked and glued locally.

[Example] A presheaf of bounded holomorphic functions that is not a sheaf. Let $X = \mathbb{C}^n$ and define

$$\mathcal{B}(U) := \{f \in \mathcal{O}(U) : f \text{ is bounded on } U\}.$$

Restriction preserves boundedness, and the identity and composition laws for restrictions are immediate. Hence $\mathcal{B}$ is a presheaf.

It is not a sheaf. Cover $\mathbb{C}^n$ by

$$U_m = \{z \in \mathbb{C}^n : \|z\| < m\}, \quad m \geq 1,$$

and define

$$f_m(z) = z_1 \quad \text{on } U_m.$$

Each f_m is bounded on U_m , and the sections agree on all overlaps. Their unique holomorphic gluing is the function

$$f(z) = z_1$$

on $\mathbb{C}^n$, which is unbounded and therefore does not belong to $\mathcal{B}(\mathbb{C}^n)$. Thus compatible locally bounded sections need not have a uniformly bounded global gluing.

[Example] A constant presheaf that is not a sheaf. Let G be a nontrivial group and define

$$\mathcal{G}^{\text{pre}}(U) = G$$

for every open set $U \subset X$, including $U = \emptyset$, and let every restriction map be

$$\rho_V^U = \text{Id}_G.$$

These assignments satisfy the identity and composition axioms, so they form a presheaf. They do not form a sheaf: applying the sheaf axiom to the empty cover of $\emptyset$ says that there must be a unique section over $\emptyset$. Equivalently, a sheaf of groups must satisfy

$$\mathcal{F}(\emptyset) = \{e\}.$$

Here $\mathcal{G}^{\text{pre}}(\emptyset) = G$, which has more than one element.

[Remark] The associated constant sheaf assigns to U the locally constant maps $U \rightarrow G$. Such a map may take different values on different connected components, which is exactly the freedom missing from the constant presheaf.

Day 5

18 Further examples and operations on sheaves

[Example] Let $\pi : E \rightarrow X$ be a holomorphic vector bundle. Its sheaf of holomorphic sections is

$$\mathcal{E}(U) := \Gamma(U, E),$$

with the usual restriction maps. The sheaf axioms follow because holomorphic sections that agree on overlaps glue uniquely.

[Example] Let G be a group. The assignment

$$U \mapsto \{\text{locally constant maps } U \rightarrow G\}$$

is the constant sheaf with value G . If U is connected, every such map is constant, so its sections over U may be identified with G .

[Example] Let $f : X \rightarrow Y$ be a smooth proper submersion and let $\underline{\mathbb{Q}}_X$ denote the constant sheaf on X . For $j \geq 0$, consider the presheaf

$$U \mapsto H^j(f^{-1}(U), \mathbb{Q}), \quad U \subset Y.$$

For an inclusion $V \subset U$, one has $f^{-1}(V) \subset f^{-1}(U)$, and restriction of cohomology classes gives the presheaf restriction map

$$H^j(f^{-1}(U), \mathbb{Q}) \longrightarrow H^j(f^{-1}(V), \mathbb{Q}).$$

These maps satisfy the identity and composition laws because ordinary restriction of cochains does. The associated sheaf is denoted by

$$R^j f_* \underline{\mathbb{Q}}_X.$$

We now compute its stalk. Fix $y \in Y$ and write

$$X_y := f^{-1}(y).$$

Because f is a proper submersion, Ehresmann's theorem says that f is locally a smooth fibre bundle. Hence there is a sufficiently small contractible neighbourhood U of y and a diffeomorphism over U

$$f^{-1}(U) \cong U \times X_y.$$

The projection $U \times X_y \rightarrow X_y$ is a homotopy equivalence because U is contractible. Therefore

$$H^j(f^{-1}(U), \mathbb{Q}) \cong H^j(U \times X_y, \mathbb{Q}) \cong H^j(X_y, \mathbb{Q}).$$

The same conclusion holds after shrinking U to a smaller contractible trivializing neighbourhood, and the resulting identifications are compatible with restriction. Taking the direct limit over all neighbourhoods of y therefore gives

$$(R^j f_* \underline{\mathbb{Q}}_X)_y \cong \varinjlim_{y \in U} H^j(f^{-1}(U), \mathbb{Q}) \cong H^j(X_y, \mathbb{Q}).$$

Thus $R^j f_* \underline{\mathbb{Q}}_X$ records how the cohomology of the fibres varies over the base. On every trivializing neighbourhood it is a constant sheaf with fibre $H^j(X_y, \mathbb{Q})$, so it is locally constant. It need not be globally constant: continuation around a loop in Y can act nontrivially on $H^j(X_y, \mathbb{Q})$, producing monodromy.

Kernels and images

Let

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of sheaves. Define

$$(\ker \varphi)(U) := \ker(\mathcal{F}(U) \xrightarrow{\varphi_U} \mathcal{G}(U)).$$

[Proposition] The presheaf $\ker \varphi$ is a sheaf.

[Proof] First, it is a presheaf. If $V \subset U$ and $s \in (\ker \varphi)(U)$, naturality of φ gives

$$\varphi_V(s|_V) = \varphi_U(s)|_V = 0,$$

so $s|_V \in (\ker \varphi)(V)$.

Now let $U = \bigcup_i U_i$ and suppose that

$$s_i \in (\ker \varphi)(U_i)$$

agree on all overlaps. Since $\mathcal{F}$ is a sheaf, there is a unique section $s \in \mathcal{F}(U)$ satisfying $s|_{U_i} = s_i$. For every i ,

$$\varphi_U(s)|_{U_i} = \varphi_{U_i}(s_i) = 0.$$

The locality axiom for $\mathcal{G}$ now implies $\varphi_U(s) = 0$. Hence $s \in (\ker \varphi)(U)$, proving existence of the required gluing. Its uniqueness follows from uniqueness of gluing in $\mathcal{F}$. Therefore $\ker \varphi$ is a sheaf.

By contrast, the sectionwise image

$$(\mathrm{im}_{\mathrm{pre}} \varphi)(U) := \mathrm{im}(\mathcal{F}(U) \xrightarrow{\varphi_U} \mathcal{G}(U))$$

is in general only a presheaf.

[Example] Let $X = \mathbb{C}^*$ and consider the exponential morphism

$$\exp : \mathcal{O}_X \longrightarrow \mathcal{O}_X^*, \quad f \longmapsto e^f,$$

where $\mathcal{O}_X^*$ is the sheaf of nowhere-vanishing holomorphic functions. Cover X by the two slit planes

$$U_1 = X \setminus (-\infty, 0), \quad U_2 = X \setminus (0, \infty).$$

On each U_i there is a holomorphic branch $\log_i z$, and hence

$$z = e^{\log_i z} \in (\text{im}_{\text{pre exp}})(U_i).$$

These local image sections are restrictions of the same global section $z \in \mathcal{O}_X^*(X)$. If the presheaf image were a sheaf, they would glue to an element of $(\text{im}_{\text{pre exp}})(X)$, so there would be a global holomorphic function f on $\mathbb{C}^*$ with $e^f = z$. Such an f would be a global holomorphic logarithm of z , which does not exist. Therefore the sectionwise image is not a sheaf.

19 Sheafification and exact sequences

[Theorem] Sheafification. For every presheaf $\mathcal{F}$ on X , there is a sheaf $\mathcal{F}^+$ and a morphism

$$\iota : \mathcal{F} \longrightarrow \mathcal{F}^+$$

with the following universal property: for every sheaf $\mathcal{G}$ and every morphism of presheaves $\alpha : \mathcal{F} \rightarrow \mathcal{G}$, there is a unique morphism of sheaves $\tilde{\alpha} : \mathcal{F}^+ \rightarrow \mathcal{G}$ such that

$$\alpha = \tilde{\alpha} \circ \iota.$$

Construction of the sheafification

For an open set $U \subset X$, let $\mathcal{F}^+(U)$ consist of maps

$$s : U \longrightarrow \coprod_{p \in U} \mathcal{F}_p$$

such that:

1. $s(p) \in \mathcal{F}_p$ for every $p \in U$;
2. every $p \in U$ has an open neighbourhood $V \subset U$ and a section $t \in \mathcal{F}(V)$ for which

$$s(q) = t_q \quad (q \in V),$$

where t_q denotes the germ of t at q .

Restrictions are obtained by restricting these maps. The morphism ι_U sends a section $t \in \mathcal{F}(U)$ to its germ-valued function

$$\tilde{t} : U \longrightarrow \coprod_{p \in U} \mathcal{F}_p, \quad \tilde{t}(p) = t_p.$$

The local representation condition makes $\mathcal{F}^+$ a sheaf, and the natural maps on stalks are isomorphisms:

$$\mathcal{F}_p \cong \mathcal{F}_p^+.$$

The universal morphism $\tilde{\alpha}$ is defined locally by applying α to a local representative. The sheaf axioms show that these local definitions glue and that the result is unique.

[Definition] The image sheaf of a morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is the sheafification of its sectionwise image presheaf:

$$\text{Im } \varphi := (\text{im}_{\text{pre}} \varphi)^+.$$

[Definition] A sequence of sheaves and sheaf morphisms

$$\cdots \longrightarrow \mathcal{F}_{i-1} \xrightarrow{d_{i-1}} \mathcal{F}_i \xrightarrow{d_i} \mathcal{F}_{i+1} \longrightarrow \cdots$$

is exact at $\mathcal{F}_i$ if

$$\ker d_i = \text{Im } d_{i-1}.$$

In particular, a sequence

$$0 \longrightarrow \mathcal{F}_1 \xrightarrow{f} \mathcal{F}_2 \xrightarrow{g} \mathcal{F}_3 \longrightarrow 0$$

is short exact when it is exact at all three nonzero terms.

[Proposition] A sequence of sheaves is exact if and only if the induced sequence on every stalk is exact. Thus the preceding short sequence is exact precisely when

$$0 \longrightarrow (\mathcal{F}_1)_p \xrightarrow{f_p} (\mathcal{F}_2)_p \xrightarrow{g_p} (\mathcal{F}_3)_p \longrightarrow 0$$

is exact for every $p \in X$.

[Proof] Consider two consecutive maps $\mathcal{F}_{i-1} \xrightarrow{d_{i-1}} \mathcal{F}_i \xrightarrow{d_i} \mathcal{F}_{i+1}$. Taking stalks commutes with kernels and images:

$$(\ker d_i)_p = \ker((d_i)_p), \quad (\text{Im } d_{i-1})_p = \text{im}((d_{i-1})_p).$$

Indeed, the first equality follows directly from the definition of a germ. For the second, a germ lies in $(\text{Im } d_{i-1})_p$ precisely when, after shrinking to a neighbourhood of p , it is represented by $d_{i-1}(s)$ for some local section s ; this is precisely the condition that it lie in $\text{im}((d_{i-1})_p)$.

Now the original sequence is exact at $\mathcal{F}_i$ exactly when $\ker d_i = \text{Im } d_{i-1}$. Two subsheaves are equal if and only if all their stalks are equal, so this is equivalent to

$$\ker((d_i)_p) = \text{im}((d_{i-1})_p)$$

for every $p \in X$. Applying this at every term proves the claim.

[Remark] Surjectivity of g as a sheaf morphism means local surjectivity: a section of $\mathcal{F}_3$ can be lifted near each point. It need not admit a lift over the whole open set.

20 Čech cohomology

Let $\mathfrak{U} = \{U_i\}_{i \in I}$ be an open cover of X , and let $\mathcal{F}$ be a sheaf of abelian groups. For $p \geq 0$, a Čech p -cochain is an alternating family

$$\sigma = (\sigma_{i_0 \dots i_p}), \quad \sigma_{i_0 \dots i_p} \in \mathcal{F}(U_{i_0} \cap \dots \cap U_{i_p}).$$

Alternating means that $\sigma_{i_0 \dots i_p} = 0$ when two indices coincide and

$$\sigma_{i_{\tau(0)} \dots i_{\tau(p)}} = \text{sgn}(\tau) \sigma_{i_0 \dots i_p}$$

for every permutation τ . The group of such cochains is denoted by $C^p(\mathfrak{U}, \mathcal{F})$.

The Čech coboundary

$$\delta : C^p(\mathfrak{U}, \mathcal{F}) \longrightarrow C^{p+1}(\mathfrak{U}, \mathcal{F})$$

is defined by

$$(\delta\sigma)_{i_0 \dots i_{p+1}} = \sum_{\nu=0}^{p+1} (-1)^\nu \sigma_{i_0 \dots \widehat{i_\nu} \dots i_{p+1}}|_{U_{i_0} \cap \dots \cap U_{i_{p+1}}}.$$

For example, if $\sigma \in C^1(\mathfrak{U}, \mathcal{F})$, then

$$(\delta\sigma)_{ijk} = \sigma_{jk} - \sigma_{ik} + \sigma_{ij} = \sigma_{ij} + \sigma_{jk} + \sigma_{ki}.$$

A direct cancellation shows that $\delta^2 = 0$. The Čech cohomology of the cover is

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) := \frac{\ker(\delta : C^p \rightarrow C^{p+1})}{\text{im}(\delta : C^{p-1} \rightarrow C^p)}.$$

For $p = 0$, the cocycle condition says precisely that the local sections agree on overlaps. The sheaf axiom therefore gives

$$\check{H}^0(\mathfrak{U}, \mathcal{F}) \cong \mathcal{F}(X).$$

Thus degree-zero cohomology is independent of the cover, whereas $\check{H}^p(\mathfrak{U}, \mathcal{F})$ for $p \geq 1$ may depend on it.

If $\mathfrak{V} = \{V_a\}$ refines $\mathfrak{U}$, a choice of refinement map produces a cochain map

$$C^\bullet(\mathfrak{U}, \mathcal{F}) \longrightarrow C^\bullet(\mathfrak{V}, \mathcal{F})$$

and hence a map on cohomology. The induced map is independent of the chosen refinement map. Passing over all open covers gives

$$\check{H}^p(X, \mathcal{F}) := \varinjlim_{\mathfrak{U}} \check{H}^p(\mathfrak{U}, \mathcal{F}).$$

21 Fundamental results in sheaf cohomology

[Theorem] A short exact sequence of sheaves

$$0 \longrightarrow \mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H} \longrightarrow 0$$

induces, by the snake lemma, a long exact sequence in cohomology:

$$\begin{aligned} 0 \longrightarrow H^0(X, \mathcal{F}) \longrightarrow H^0(X, \mathcal{G}) \longrightarrow H^0(X, \mathcal{H}) \\ \longrightarrow H^1(X, \mathcal{F}) \longrightarrow H^1(X, \mathcal{G}) \longrightarrow H^1(X, \mathcal{H}) \longrightarrow \cdots \end{aligned}$$

[Proposition] Acyclic-resolution proposition. Suppose

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G}_0 \xrightarrow{d_0} \mathcal{G}_1 \xrightarrow{d_1} \cdots \xrightarrow{d_{m-1}} \mathcal{G}_m \longrightarrow 0$$

is an exact sequence of sheaves and

$$H^q(X, \mathcal{G}_j) = 0 \quad (q \geq 1, 0 \leq j \leq m).$$

Then the cohomology of $\mathcal{F}$ is computed by the complex of global sections:

$$H^q(X, \mathcal{F}) \cong H^q \left(0 \longrightarrow \mathcal{G}_0(X) \xrightarrow{d_0} \mathcal{G}_1(X) \xrightarrow{d_1} \cdots \xrightarrow{d_{m-1}} \mathcal{G}_m(X) \longrightarrow 0 \right).$$

[Proof] Put

$$\mathcal{K}_j := \ker(d_j : \mathcal{G}_j \rightarrow \mathcal{G}_{j+1}),$$

where $d_m : \mathcal{G}_m \rightarrow 0$. Thus $\mathcal{K}_0 \cong \mathcal{F}$. Moreover, exactness gives $\text{Im } d_j = \mathcal{K}_{j+1}$, so for $0 \leq j < m$ we have short exact sequences

$$0 \longrightarrow \mathcal{K}_j \longrightarrow \mathcal{G}_j \longrightarrow \mathcal{K}_{j+1} \longrightarrow 0.$$

For $r \geq 1$, the corresponding long exact sequence contains

$$H^r(X, \mathcal{G}_j) \longrightarrow H^r(X, \mathcal{K}_{j+1}) \xrightarrow{\delta_j} H^{r+1}(X, \mathcal{K}_j) \longrightarrow H^{r+1}(X, \mathcal{G}_j).$$

The two outer groups vanish by hypothesis. Hence the connecting map is an isomorphism:

$$H^r(X, \mathcal{K}_{j+1}) \xrightarrow[\delta_j]{\sim} H^{r+1}(X, \mathcal{K}_j).$$

Let $C^\bullet$ be the complex of global sections. For $1 \leq q \leq m$, apply cohomology to $0 \rightarrow \mathcal{K}_{q-1} \rightarrow \mathcal{G}_{q-1} \rightarrow \mathcal{K}_q \rightarrow 0$. The relevant part of its long exact sequence is

$$H^0(X, \mathcal{G}_{q-1}) \longrightarrow H^0(X, \mathcal{K}_q) \longrightarrow H^1(X, \mathcal{K}_{q-1}) \longrightarrow H^1(X, \mathcal{G}_{q-1}).$$

Here $H^0(X, \mathcal{A}) = \mathcal{A}(X)$ for every sheaf $\mathcal{A}$, which is why the first two terms may also be written as $\mathcal{G}_{q-1}(X)$ and $\mathcal{K}_q(X)$. The last group is zero by the hypothesis, so this segment ends in 0.

Now $\mathcal{K}_q(X) = \ker(d_q : \mathcal{G}_q(X) \rightarrow \mathcal{G}_{q+1}(X))$. The connecting map $\mathcal{K}_q(X) \rightarrow H^1(X, \mathcal{K}_{q-1})$ is surjective, and exactness says that its kernel is $d_{q-1}(\mathcal{G}_{q-1}(X))$. Hence the first isomorphism theorem gives

$$\begin{aligned} H^q(C^\bullet) &= \frac{\mathcal{K}_q(X)}{d_{q-1}(\mathcal{G}_{q-1}(X))} \\ &\cong H^1(X, \mathcal{K}_{q-1}). \end{aligned}$$

Applying the connecting isomorphisms successively gives the dimension-shifting chain

$$H^1(X, \mathcal{K}_{q-1}) \cong H^2(X, \mathcal{K}_{q-2}) \cong \cdots \cong H^q(X, \mathcal{K}_0) \cong H^q(X, \mathcal{F}).$$

For $q = 0$, $H^0(C^\bullet) = \ker(\mathcal{G}_0(X) \rightarrow \mathcal{G}_1(X)) = \mathcal{K}_0(X)$, which is $H^0(X, \mathcal{F})$.

Finally, if $q > m$, then $H^q(C^\bullet) = 0$, while dimension shifting gives $H^q(X, \mathcal{F}) \cong H^{q-m}(X, \mathcal{K}_m) = H^{q-m}(X, \mathcal{G}_m) = 0$.

Acyclic, fine, and flabby sheaves

[Definition] A sheaf $\mathcal{F}$ is acyclic if

$$H^q(X, \mathcal{F}) = 0 \quad (q \geq 1).$$

[Definition] A sheaf $\mathcal{F}$ is fine if, for every locally finite open cover $\{U_i\}_{i \in I}$ of X , there are sheaf endomorphisms

$$\eta_i : \mathcal{F} \longrightarrow \mathcal{F}$$

such that

$$\text{supp}(\eta_i) \subset U_i \quad \text{and} \quad \sum_{i \in I} \eta_i = \text{Id}_{\mathcal{F}}.$$

The sum is locally finite. This is the sheaf-theoretic analogue of a partition of unity.

[Theorem] Fine sheaves are acyclic.

[Example] On a smooth manifold, the sheaves $\mathcal{A}^p$ of smooth complex-valued p -forms and the sheaves $\mathcal{A}^{p,q}$ of smooth (p, q) -forms are fine. The endomorphisms η_i are multiplication by a smooth partition of unity subordinate to the cover. In contrast, the sheaf $\mathcal{O}_X$ of holomorphic functions is generally not fine, because holomorphic partitions of unity usually do not exist.

[Definition] A sheaf $\mathcal{F}$ is flabby, or flasque, if every restriction homomorphism is surjective:

$$\rho_V^U : \mathcal{F}(U) \longrightarrow \mathcal{F}(V)$$

is onto whenever $V \subset U$ are open.

[Theorem] Flabby sheaves are acyclic.

22 The de Rham and Dolbeault theorems

[Theorem] de Rham theorem. Let X be a smooth manifold and let $\mathbb{C}_X$ be its constant sheaf. Then

$$H^p(X, \mathbb{C}_X) \cong H_{\text{dR}}^p(X; \mathbb{C}) = H^p\left(\mathcal{A}^0(X) \xrightarrow{d} \mathcal{A}^1(X) \xrightarrow{d} \dots\right).$$

[Definition] For a complex manifold X , Ω_X^p denotes the sheaf of holomorphic p -forms. Thus, for every open set $U \subset X$,

$$\Omega_X^p(U) = \{\text{holomorphic } p\text{-forms on } U\}.$$

In local holomorphic coordinates, its sections have the form $\sum_{|I|=p} f_I(z) dz_I$ with holomorphic coefficients f_I . Hence Ω_X^p is a subsheaf of the sheaf $\mathcal{A}^{p,0}$ of smooth $(p,0)$ -forms, and

$$\Omega_X^p = \ker\left(\bar{\partial} : \mathcal{A}^{p,0} \longrightarrow \mathcal{A}^{p,1}\right).$$

[Theorem] Dolbeault theorem. Let X be a complex manifold and $p \geq 0$. Then

$$H^q(X, \Omega_X^p) \cong H_{\bar{\partial}}^{p,q}(X) = H^q\left(\mathcal{A}^{p,0}(X) \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1}(X) \xrightarrow{\bar{\partial}} \dots\right).$$

[Proof] For the de Rham theorem, consider the sequence of sheaves

$$0 \longrightarrow \mathbb{C}_X \longrightarrow \mathcal{A}^0 \xrightarrow{d} \mathcal{A}^1 \xrightarrow{d} \mathcal{A}^2 \xrightarrow{d} \dots$$

It is exact by the Poincaré lemma. Since each $\mathcal{A}^p$ is fine and therefore acyclic, this is an acyclic resolution of $\mathbb{C}_X$. The acyclic-resolution proposition computes its sheaf cohomology using global sections, giving

$$H^p(X, \mathbb{C}_X) \cong H_{\text{dR}}^p(X; \mathbb{C}) = H^p\left(\mathcal{A}^0(X) \xrightarrow{d} \mathcal{A}^1(X) \xrightarrow{d} \dots\right).$$

For the Dolbeault theorem, the Dolbeault lemma makes

$$0 \longrightarrow \Omega_X^p \longrightarrow \mathcal{A}^{p,0} \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1} \xrightarrow{\bar{\partial}} \mathcal{A}^{p,2} \xrightarrow{\bar{\partial}} \dots$$

an exact sequence. Since the sheaves $\mathcal{A}^{p,q}$ are fine and therefore acyclic, this is an acyclic resolution of Ω_X^p . Applying the same proposition gives

$$H^q(X, \Omega_X^p) \cong H_{\bar{\partial}}^{p,q}(X) = H^q\left(\mathcal{A}^{p,0}(X) \xrightarrow{\bar{\partial}} \mathcal{A}^{p,1}(X) \xrightarrow{\bar{\partial}} \dots\right).$$

Day 6

23 Applications of sheaf cohomology

Acyclic resolutions

[Remark] Recall that if

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G}_0 \longrightarrow \mathcal{G}_1 \longrightarrow \cdots$$

is an exact resolution and every $\mathcal{G}_j$ is acyclic, meaning $H^q(X, \mathcal{G}_j) = 0$ for $q > 0$, then the acyclic-resolution proposition gives

$$H^q(X, \mathcal{F}) \cong H^q(\Gamma(X, \mathcal{G}_\bullet)),$$

where $\Gamma(X, \mathcal{G}_\bullet)$ is the complex

$$\Gamma(X, \mathcal{G}_0) \longrightarrow \Gamma(X, \mathcal{G}_1) \longrightarrow \cdots$$

Constant sheaves and singular cohomology

[Theorem] Comparison theorem. If X is paracompact and locally contractible, as is any manifold, then

$$H^q(X, \mathbb{Z}_X) \cong H_{\text{sing}}^q(X; \mathbb{Z}).$$

Thus the sheaf cohomology of the constant sheaf recovers the ordinary integral cohomology of X from algebraic topology.

The Picard group

[Definition] Let $\mathcal{O}_X^*$ denote the sheaf of nowhere-vanishing holomorphic functions on a complex manifold X . The **Picard group** is

$$\text{Pic}(X) := H^1(X, \mathcal{O}_X^*).$$

The group operation corresponds to tensor product of holomorphic line bundles.

[Proposition] There is a natural bijection

$$H^1(X, \mathcal{O}_X^*) \longleftrightarrow \{\text{holomorphic line bundles on } X\} / \cong.$$

[Proof] Let $L \rightarrow X$ be a holomorphic line bundle. Choose an open cover $\{U_i\}$ and nowhere-vanishing holomorphic frames e_i of $L|_{U_i}$. On an overlap write

$$e_j = g_{ij}e_i, \quad g_{ij} \in \mathcal{O}_X^*(U_i \cap U_j).$$

On triple overlaps, $g_{ij}g_{jk} = g_{ik}$, so $\{g_{ij}\}$ is a Čech 1-cocycle. Replacing e_i by h_ie_i , with

$h_i \in \mathcal{O}_X^*(U_i)$, changes g_{ij} by a Cech coboundary. Hence the cohomology class depends only on L .

Conversely, a cocycle $\{g_{ij}\}$ glues the trivial bundles $U_i \times \mathbb{C}$ using the relation

$$(x, v)_j \sim (x, g_{ij}(x)v)_i.$$

Cohomologous cocycles give isomorphic bundles. Under this correspondence, multiplication of cocycles gives tensor product of line bundles.

Divisors

[Definition] A **prime divisor** on X is an irreducible reduced analytic subvariety of complex codimension one. Locally it is defined by one reduced irreducible holomorphic function:

$$D \cap U_i = \{f_i = 0\}, \quad f_i \in \mathcal{O}_X(U_i).$$

[Definition] A **divisor** is a locally finite formal integer combination of prime divisors,

$$D = \sum_{\alpha} n_{\alpha} D_{\alpha}, \quad n_{\alpha} \in \mathbb{Z}.$$

[Definition] If g is a nonzero meromorphic function, its **principal divisor** is

$$(g) = \sum_D \text{ord}_D(g) D,$$

where $\text{ord}_D(g)$ is the order of vanishing, negative for a pole. Two divisors are **linearly equivalent**, written $D_1 \sim D_2$, if

$$D_1 - D_2 = (g)$$

for some nonzero meromorphic function g .

[Example] On $\mathbb{P}^n$, the meromorphic function $g = z_0/z_1$ has

$$(g) = \{z_0 = 0\} - \{z_1 = 0\}.$$

Thus any two hyperplanes are linearly equivalent.

[Proposition] A divisor D determines a holomorphic line bundle $\mathcal{O}_X(D)$. Linearly equivalent divisors determine isomorphic line bundles.

[Proof] Choose local meromorphic defining functions f_i for D on U_i . Then $g_{ij} := f_i/f_j \in \mathcal{O}_X^*(U_i \cap U_j)$ is a cocycle. Glue local frames by $e_j = g_{ij}e_i$; with this convention, the local expressions $f_i e_i$ agree and define a global meromorphic section whose divisor is D . Replacing the f_i by another set of local defining functions changes the cocycle by a coboundary. If $D_1 - D_2 = (g)$, multiplication by g gives $\mathcal{O}_X(D_1) \cong \mathcal{O}_X(D_2)$.

Conversely, a nonzero meromorphic section s of a holomorphic line bundle determines $(s) =$

$$(s)_0 - (s)_\infty.$$

When such meromorphic sections exist, this recovers the line bundle from its divisor class.

The exponential sequence and the first Chern class

The exponential sequence of sheaves is

$$0 \longrightarrow \mathbb{Z}_X \longrightarrow \mathcal{O}_X \xrightarrow{\exp(2\pi i \cdot)} \mathcal{O}_X^* \longrightarrow 0.$$

Its long exact sequence contains

$$H^1(X, \mathcal{O}_X) \longrightarrow \text{Pic}(X) \xrightarrow{c_1} H^2(X, \mathbb{Z}_X) \longrightarrow H^2(X, \mathcal{O}_X).$$

[Definition] The connecting homomorphism assigns to a holomorphic line bundle L its **first Chern class**

$$c_1(L) \in H^2(X, \mathbb{Z}_X)$$

The subgroup

$$\text{Pic}^0(X) := \ker c_1$$

consists of topologically trivial holomorphic line bundles, and exactness gives

$$\text{Pic}^0(X) \cong \frac{H^1(X, \mathcal{O}_X)}{\text{im } H^1(X, \mathbb{Z}_X)}.$$

For compact Kahler X , $\text{Pic}^0(X)$ is a compact complex torus.

24 Differential geometry of vector bundles

Hermitian metrics

[Definition] Let $E \rightarrow X$ be a holomorphic vector bundle of rank r . A **Hermitian metric** on E is a smoothly varying family of Hermitian inner products

$$h_x : E_x \times E_x \longrightarrow \mathbb{C}.$$

In a local frame $e = (e_1, \dots, e_r)$, the metric is represented by the positive-definite Hermitian matrix

$$H = (h_{\alpha\bar{\beta}}), \quad h_{\alpha\bar{\beta}} = h(e_\alpha, e_\beta).$$

[Remark] In tensorial language, using the convention that h is complex-linear in its first argument and conjugate-linear in its second, a Hermitian metric is a positive-definite smooth section $h \in \Gamma(X, E^* \otimes \overline{E}^*)$, locally written $h = h_{\alpha\bar{\beta}} e^\alpha \otimes \overline{e}^\beta$. Equivalently, it is the bundle isomorphism $h^\flat : E \rightarrow \overline{E}^*$, $h^\flat(v) = h(v, \cdot)$, satisfying Hermitian symmetry and positivity. Hermitian metrics exist: choose local metrics in trivializations and combine them using a smooth partition of unity.

Connections and curvature

[Definition] A **connection** on a smooth complex vector bundle $E \rightarrow X$ is a $\mathbb{C}$ -linear map

$$D : C^\infty(X, E) \longrightarrow \mathcal{A}^1(X, E)$$

satisfying the Leibniz rule

$$D(fs) = df \otimes s + fDs.$$

In a local frame $e = (e_1, \dots, e_r)$, write

$$De = e\theta,$$

where $\theta = (\theta_\beta^\alpha)$ is a matrix of one-forms, called the **connection matrix**. If $s = eu$ for a column vector u of functions, then

$$Ds = e(du + \theta u).$$

[Definition] Extend D to E -valued forms by

$$D(\varphi \otimes s) = d\varphi \otimes s + (-1)^k \varphi \wedge Ds, \quad \varphi \in \mathcal{A}^k(X).$$

The **curvature** is

$$\Theta := D^2 \in \mathcal{A}^2(X, \text{End } E).$$

Unlike D , the curvature is tensorial: $\Theta(fs) = f\Theta(s)$. In a local frame,

$$\Theta = d\theta + \theta \wedge \theta.$$

Under a change of frame $\tilde{e} = eA$, one has

$$\tilde{\theta} = A^{-1}\theta A + A^{-1}dA, \quad \tilde{\Theta} = A^{-1}\Theta A.$$

Thus the connection matrix is not tensorial, whereas the curvature matrix transforms as an endomorphism-valued two-form.

25 The Chern connection

[Theorem] Chern connection. Let $(E, h) \rightarrow X$ be a holomorphic Hermitian vector bundle. There is a unique connection D such that:

1. D is compatible with the metric,

$$dh(s, t) = h(Ds, t) + h(s, Dt);$$

1. the $(0, 1)$ -part of D is the holomorphic structure,

$$D^{0,1} = \bar{\partial}_E.$$

Equivalently, if s is holomorphic, then Ds is an E -valued $(1, 0)$ -form.

In a local holomorphic frame with metric matrix H , the Chern connection and its curvature are

$$\theta = H^{-1}\partial H, \quad \Theta = \bar{\partial}(H^{-1}\partial H).$$

In particular, for a line bundle with a local holomorphic frame e and $h(e, e) = h > 0$,

$$\theta = \partial \log h, \quad \Theta = \bar{\partial} \partial \log h = -\partial \bar{\partial} \log h.$$

The tautological bundle on projective space

[Example] Equip the tautological line bundle $\mathcal{O}_{\mathbb{P}^n}(-1)$ with the metric induced by the standard Hermitian metric on $\mathbb{C}^{n+1}$. On the affine chart $U_0 = \{z_0 \neq 0\}$, with coordinates $w_j = z_j/z_0$, the standard holomorphic frame has squared norm

$$h = 1 + \sum_{j=1}^n |w_j|^2.$$

Therefore

$$\Theta_{\mathcal{O}(-1)} = -\partial \bar{\partial} \log \left(1 + \sum_{j=1}^n |w_j|^2 \right),$$

and

$$c_1(\mathcal{O}(-1), h) = \frac{i}{2\pi} \Theta_{\mathcal{O}(-1)} = -\omega_{\text{FS}}.$$

26 Chern-Weil theory

[Definition] For a connection D with curvature Θ , define the total Chern form by

$$c(E, D) := \det \left(I + \frac{i}{2\pi} \Theta \right) = 1 + c_1(E, D) + \cdots + c_r(E, D),$$

where $c_k(E, D)$ is a differential form of degree $2k$.

[Theorem] Chern-Weil theorem. The forms $c_k(E, D)$ are closed, and their de Rham cohomology classes are independent of the chosen connection. In particular, if D_0 and D_1 are two connections, then

$$c_k(E, D_1) - c_k(E, D_0)$$

is exact. For a holomorphic Hermitian bundle, the classes computed from the Chern connection are therefore independent of the Hermitian metric.

[Definition] The resulting class

$$c_k(E) := [c_k(E, D)] \in H_{\text{dR}}^{2k}(X; \mathbb{C})$$

is the k -th **Chern class** of E . For a line bundle, this differential-geometric first Chern class is the image in de Rham cohomology of the integral class obtained from the exponential sequence.

27 Properties of Chern classes

[Proposition] Naturality. If $f : X \rightarrow Y$ is smooth and $E \rightarrow Y$ is a complex vector bundle, then

$$c_k(f^*E) = f^*c_k(E).$$

[Proposition] Whitney product formula. For complex vector bundles E and F on X ,

$$c(E \oplus F) = c(E) \wedge c(F).$$

More generally, if

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

is a short exact sequence of complex vector bundles, then

$$c(E) = c(E') \wedge c(E'').$$

Day 7

28 Hermitian and Kähler metrics

Let (X, J) be a complex manifold and let g be a Riemannian metric on the underlying real tangent bundle $T_{\mathbb{R}}X$. We say that g is J -invariant if

$$g(JX, JY) = g(X, Y)$$

for all real tangent vectors X, Y . Extend g complex-bilinearly to the complexified tangent bundle

$$T_{\mathbb{C}}X = T_{\mathbb{R}}X \otimes_{\mathbb{R}} \mathbb{C} = T^{1,0}X \oplus T^{0,1}X.$$

If g is J -invariant, then

$$g^{\mathbb{C}}(T^{1,0}X, T^{1,0}X) = 0, \quad g^{\mathbb{C}}(T^{0,1}X, T^{0,1}X) = 0.$$

Indeed, if $u, v \in T^{1,0}X$, then $Ju = iu$ and $Jv = iv$, so

$$g^{\mathbb{C}}(u, v) = g^{\mathbb{C}}(Ju, Jv) = g^{\mathbb{C}}(iu, iv) = -g^{\mathbb{C}}(u, v).$$

[Definition] The associated Hermitian metric on $T^{1,0}X$ is

$$h(u, v) = g^{\mathbb{C}}(u, \bar{v}), \quad u, v \in T^{1,0}X.$$

It is complex-linear in the first variable, conjugate-linear in the second, and positive definite.

Conversely, a Hermitian metric h on $T^{1,0}X$ determines a J -invariant Riemannian metric by

$$g(X, Y) = 2 \operatorname{Re} h(X^{1,0}, Y^{1,0}), \quad X^{1,0} = \frac{1}{2}(X - iJX).$$

Thus J -invariant Riemannian metrics on $T_{\mathbb{R}}X$ are equivalent to Hermitian metrics on $T^{1,0}X$.

[Definition] The **fundamental form** of a Hermitian manifold (X, J, g) is

$$\omega(X, Y) = g(JX, Y).$$

The J -invariance of g implies

$$g(JX, Y) = -g(X, JY),$$

so ω is a real 2-form. In holomorphic coordinates $z^1, \dots, z^n$, writing

$$g_{i\bar{j}} = g^{\mathbb{C}}\left(\frac{\partial}{\partial z^i}, \frac{\partial}{\partial \bar{z}^j}\right),$$

we have

$$\omega = ig_{i\bar{j}} dz^i \wedge d\bar{z}^j.$$

[Definition] A Hermitian metric is **Kähler** if

$$d\omega = 0.$$

In this case ω is called the **Kähler form**. Since ω is nondegenerate, a Kähler form is in particular a symplectic form.

Because ω has type $(1, 1)$,

$$d\omega = 0$$

is equivalent to

$$\frac{\partial g_{i\bar{j}}}{\partial z^k} = \frac{\partial g_{k\bar{j}}}{\partial z^i}, \quad \frac{\partial g_{i\bar{j}}}{\partial \bar{z}^k} = \frac{\partial g_{i\bar{k}}}{\partial \bar{z}^j}.$$

[Proposition] Let (X, J, g) be Kähler and let $p \in X$. There exist holomorphic coordinates centered at p in which

$$g_{i\bar{j}}(p) = \delta_{ij}, \quad \frac{\partial g_{i\bar{j}}}{\partial z^k}(p) = \frac{\partial g_{i\bar{j}}}{\partial \bar{z}^k}(p) = 0.$$

Consequently,

$$g_{i\bar{j}}(z) = \delta_{ij} + O(|z|^2).$$

[Proof] A complex-linear change of coordinates first makes $g_{i\bar{j}}(p) = \delta_{ij}$. The Kähler identities imply that $\partial_k g_{i\bar{j}}(p)$ is symmetric in i and k . A holomorphic quadratic change of coordinates can therefore cancel all holomorphic first derivatives at p ; their conjugates vanish as well. This gives the stated expansion.

[Corollary] A Hermitian metric g is Kähler if and only if its Levi-Civita connection satisfies

$$\nabla J = 0.$$

[Proof] If g is Kähler, choose holomorphic normal coordinates at an arbitrary point p . At p the Christoffel symbols vanish and J has constant coordinate coefficients, so $(\nabla J)_p = 0$. Hence $\nabla J = 0$ everywhere.

Conversely, if $\nabla J = 0$, then $\nabla g = 0$ gives $\nabla \omega = 0$. Since the Levi-Civita connection is torsion-free,

$$d\omega = \text{Alt}(\nabla \omega) = 0,$$

so g is Kähler.

29 The Fubini–Study metric and Kähler submanifolds

Let $[z] \in \mathbb{CP}^n$, where $0 \neq z \in \mathbb{C}^{n+1}$. Using the standard Hermitian inner product on $\mathbb{C}^{n+1}$, the tangent space at $[z]$ may be represented by the horizontal subspace

$$z^\perp = \{u \in \mathbb{C}^{n+1} : \langle u, z \rangle = 0\}.$$

The induced Hermitian metric on these horizontal representatives is the **Fubini–Study metric**. Its Kähler form is

$$\omega_{\text{FS}} = \frac{i}{2\pi} \partial \bar{\partial} \log |z|^2$$

on $\mathbb{C}^{n+1} \setminus \{0\}$, viewed as the pullback of a form on $\mathbb{CP}^n$.

On the affine chart

$$U_0 = \{[z] : z_0 \neq 0\}, \quad w^i = \frac{z_i}{z_0},$$

we obtain

$$\omega_{\text{FS}} = \frac{i}{2\pi} \partial \bar{\partial} \log \left(1 + \sum_{i=1}^n |w^i|^2 \right).$$

This expression agrees with the metric induced from the horizontal subspaces $z^\perp$. It also shows immediately that

$$d\omega_{\text{FS}} = 0,$$

because $d = \partial + \bar{\partial}$, $\partial^2 = \bar{\partial}^2 = 0$, and $\partial \bar{\partial} = -\bar{\partial} \partial$.

[Remark] Although $\log |z|^2$ is not a function on projective space, the form above descends: replacing a local holomorphic lift z by fz , with f nowhere zero and holomorphic, changes the potential by $\log |f|^2$, and $\partial \bar{\partial} \log |f|^2 = 0$.

[Proposition] Let $i : Y \hookrightarrow X$ be a complex submanifold of a Kähler manifold (X, J, g) , and give Y the induced metric. Then Y is Kähler and

$$\omega_Y = i^* \omega_X.$$

[Proof] Since TY is preserved by J , the pullback i^*g is Hermitian and its fundamental form is $i^*\omega_X$. Therefore

$$d\omega_Y = d(i^*\omega_X) = i^*(d\omega_X) = 0.$$

30 The Levi-Civita and Chern connections

Let (X, J, g) be Kähler. Extend the Levi-Civita connection complex-linearly to $T_{\mathbb{C}}X$. Since $\nabla J = 0$, it preserves the splitting

$$T_{\mathbb{C}}X = T^{1,0}X \oplus T^{0,1}X$$

and hence restricts to a connection on the holomorphic tangent bundle $T^{1,0}X$.

[Theorem] On a Kähler manifold, the restriction of the complexified Levi-Civita connection to $T^{1,0}X$ is the Chern connection of the Hermitian holomorphic bundle $(T^{1,0}X, h)$.

[Proof] The restricted connection is compatible with h because the Levi-Civita connection preserves both g and J . It remains to identify its $(0,1)$ -part. For $W \in \Gamma(T^{1,0}X)$ and $\bar{Z} \in \Gamma(T^{0,1}X)$, torsion-freeness gives

$$\nabla_{\bar{Z}}W - [\bar{Z}, W] = \nabla_W\bar{Z}.$$

The right-hand side has type $(0,1)$, so taking the $(1,0)$ -part yields

$$\nabla_{\bar{Z}}W = [\bar{Z}, W]^{1,0} = \bar{\partial}_{\bar{Z}}W.$$

Thus $\nabla^{0,1} = \bar{\partial}$. The uniqueness of the Chern connection completes the proof.

In holomorphic coordinates, write

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k.$$

Metric compatibility and preservation of type give

$$\partial_i g_{j\bar{\ell}} = g^{\mathbb{C}}(\nabla_{\partial_i} \partial_j, \partial_{\bar{\ell}}) = \Gamma_{ij}^k g_{k\bar{\ell}}.$$

Therefore

$$\Gamma_{ij}^k = g^{k\bar{\ell}} \partial_i g_{j\bar{\ell}}.$$

31 Curvature and the Ricci form of a Kähler metric

For a connection D on a holomorphic vector bundle, its curvature is

$$\Theta = D^2 \in \Gamma(X, \Lambda^{1,1} T^* X \otimes \text{End}(E)).$$

For the Kähler tangent bundle, the curvature has type $(1, 1)$:

$$R(\partial_i, \partial_j) = 0, \quad R(\partial_{\bar{i}}, \partial_{\bar{j}}) = 0.$$

Equivalently,

$$R(JX, JY)Z = R(X, Y)Z.$$

Define the components

$$R_{i\bar{j}k\bar{\ell}} = g^{\mathbb{C}}(R(\partial_i, \partial_{\bar{j}})\partial_k, \partial_{\bar{\ell}}).$$

Since

$$R(\partial_i, \partial_{\bar{j}})\partial_k = -\partial_{\bar{j}}\Gamma_{ik}^p \partial_p,$$

we obtain

$$R_{i\bar{j}k\bar{\ell}} = -\partial_i \partial_{\bar{j}} g_{k\bar{\ell}} + g^{p\bar{q}}(\partial_i g_{k\bar{q}})(\partial_{\bar{j}} g_{p\bar{\ell}}).$$

In holomorphic normal coordinates at a point p , this reduces to

$$R_{i\bar{j}k\bar{\ell}}(p) = -\partial_i \partial_{\bar{j}} g_{k\bar{\ell}}(p).$$

The Ricci tensor is J -invariant:

$$\text{Ric}(JX, JY) = \text{Ric}(X, Y).$$

Consequently, the **Ricci form**

$$\rho(X, Y) = \text{Ric}(JX, Y)$$

is a real $(1, 1)$ -form. In holomorphic coordinates,

$$\rho = i R_{i\bar{j}} dz^i \wedge d\bar{z}^j, \quad R_{i\bar{j}} = g^{k\bar{\ell}} R_{i\bar{j}k\bar{\ell}}.$$

[Proposition] The Ricci form of a Kähler metric is

$$\rho = -i \partial \bar{\partial} \log \det(g_{k\bar{\ell}}).$$

In particular, $d\rho = 0$.

[Proof] If $G = (g_{k\bar{\ell}})$, then

$$\partial_i \log \det G = \text{tr}(G^{-1} \partial_i G) = \Gamma_{ik}^k.$$

Tracing the curvature identity therefore gives

$$R_{i\bar{j}} = -\partial_{\bar{j}} \Gamma_{ik}^k = -\partial_i \partial_{\bar{j}} \log \det G,$$

which proves the formula. Its right-hand side is d -closed.

[Remark] The formula is independent of holomorphic coordinates: under a coordinate change, $\det G$ is multiplied by the squared modulus of a nowhere-zero holomorphic Jacobian determinant, whose logarithm is annihilated by $\partial\bar{\partial}$.

[Corollary] With the normalization used above,

$$\left[\frac{\rho}{2\pi} \right] = c_1(T^{1,0}X).$$

Equivalently, $\rho/(2\pi)$ represents the first Chern class of the determinant line bundle

$$\det(T^{1,0}X) = \Lambda^n T^{1,0}X.$$

[Proof] Let Θ be the curvature matrix of the Chern connection on $T^{1,0}X$. In a holomorphic frame with metric matrix G ,

$$\mathrm{tr} \Theta = \mathrm{tr} \bar{\partial}(G^{-1}\partial G) = \bar{\partial}\partial \log \det G = -\partial\bar{\partial} \log \det G.$$

Consequently,

$$\rho = i \mathrm{tr} \Theta.$$

The degree-two part of the Chern-Weil form is

$$c_1(T^{1,0}X) = \left[\frac{i}{2\pi} \mathrm{tr} \Theta \right] = \left[\frac{\rho}{2\pi} \right].$$

The curvature of the induced connection on $\det(T^{1,0}X)$ is $\mathrm{tr} \Theta$, so the determinant-line-bundle formulation is equivalent.

[Remark] The bundle $\Lambda^n T^{1,0}X$ has complex rank one, and $c_1(\det E) = c_1(E)$ because the curvature of the induced determinant connection is the trace of the curvature of E .

32 Hodge theory on compact Riemannian manifolds

We now return to a compact oriented Riemannian manifold (X, g) of real dimension n . The metric on TX induces a pointwise inner product on p -forms and hence the L^2 inner product

$$(\alpha, \beta)_{L^2} := \int_X \langle \alpha, \beta \rangle_g dV_g, \quad \|\alpha\|_{L^2}^2 = (\alpha, \alpha)_{L^2}.$$

Energy minimization in a cohomology class

Let $[\alpha] \in H_{\mathrm{dR}}^p(X; \mathbb{R})$. Its representatives have the form

$$\alpha + d\eta, \quad \eta \in \mathcal{A}^{p-1}(X),$$

where $d\alpha = 0$. A canonical representative should minimize the L^2 energy among all representatives of the class.

[Definition] The **formal adjoint** of

$$d : \mathcal{A}^{p-1}(X) \longrightarrow \mathcal{A}^p(X)$$

is the operator

$$d^* : \mathcal{A}^p(X) \longrightarrow \mathcal{A}^{p-1}(X)$$

characterized by

$$(d\eta, \alpha)_{L^2} = (\eta, d^*\alpha)_{L^2}.$$

Compactness and the absence of boundary ensure that no boundary term occurs.

Suppose that α is a critical point of the L^2 energy in its cohomology class. For every $\eta \in \mathcal{A}^{p-1}(X)$, consider

$$E(t) = \|\alpha + t d\eta\|_{L^2}^2.$$

Then

$$E'(0) = 2(\alpha, d\eta)_{L^2} = 2(d^*\alpha, \eta)_{L^2}.$$

Thus $E'(0) = 0$ for every η if and only if $d^*\alpha = 0$. Since every representative under consideration is already closed, an energy-minimizing representative satisfies

$$d\alpha = 0, \quad d^*\alpha = 0.$$

The Hodge star and the formal adjoint

[Definition] The **Hodge star operator** is the bundle isomorphism

$$* : \mathcal{A}^p(X) \longrightarrow \mathcal{A}^{n-p}(X)$$

defined by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle_g dV_g$$

for all $\alpha, \beta \in \mathcal{A}^p(X)$.

If $(e^1, \dots, e^n)$ is an oriented orthonormal coframe and

$$I = (i_1 < \dots < i_p), \quad I^c = (j_1 < \dots < j_{n-p}),$$

then

$$*(e^{i_1} \wedge \dots \wedge e^{i_p}) = \text{sgn}(I, I^c) e^{j_1} \wedge \dots \wedge e^{j_{n-p}},$$

where the sign is chosen so that

$$e^{i_1} \wedge \dots \wedge e^{i_p} \wedge *(e^{i_1} \wedge \dots \wedge e^{i_p}) = dV_g.$$

[Proposition] On p -forms the Hodge star satisfies

$$*^2 = (-1)^{p(n-p)} \text{Id},$$

and it is an isometry:

$$\langle *\alpha, *\beta \rangle_g = \langle \alpha, \beta \rangle_g.$$

The formal adjoint of d is

$$d^* = (-1)^{np+n+1} * d *$$

on p -forms.

[Proof] Fix a point and an oriented orthonormal coframe $(e^1, \dots, e^n)$. For an increasing

multi-index

$$I = (i_1 < \cdots < i_p),$$

write

$$e^I = e^{i_1} \wedge \cdots \wedge e^{i_p},$$

and let $I^c = (j_1 < \cdots < j_{n-p})$ be its complementary multi-index. By definition,

$$*e^I = \varepsilon(I)e^{I^c},$$

where $\varepsilon(I)$ is the sign of the permutation

$$(i_1, \dots, i_p, j_1, \dots, j_{n-p}) \mapsto (1, \dots, n).$$

Applying $*$ again gives

$$*^2 e^I = \varepsilon(I)\varepsilon(I^c)e^I.$$

Passing from the ordered blocks (I, I^c) to (I^c, I) requires $p(n-p)$ transpositions. Therefore

$$\varepsilon(I)\varepsilon(I^c) = (-1)^{p(n-p)},$$

and hence

$$*^2 = (-1)^{p(n-p)} \text{Id}$$

on p -forms.

The forms e^I constitute an orthonormal basis of $\Lambda^p T_x^* X$, and $*$ sends each e^I to the signed orthonormal basis vector $\varepsilon(I)e^{I^c}$. It therefore preserves the pointwise inner product:

$$\langle * \alpha, * \beta \rangle_g = \langle \alpha, \beta \rangle_g.$$

It remains to compute the formal adjoint. Let $\eta \in \mathcal{A}^{p-1}(X)$ and $\alpha \in \mathcal{A}^p(X)$. Since X is compact and has no boundary, Stokes' theorem gives

$$\begin{aligned} 0 &= \int_X d(\eta \wedge * \alpha) \\ &= \int_X d\eta \wedge * \alpha + (-1)^{p-1} \int_X \eta \wedge d(* \alpha). \end{aligned}$$

Thus

$$(d\eta, \alpha)_{L^2} = \int_X d\eta \wedge * \alpha = (-1)^p \int_X \eta \wedge d(* \alpha).$$

Define, on p -forms,

$$\delta := (-1)^{np+n+1} * d *.$$

The form $d(* \alpha)$ has degree $n-p+1$. Using the already proved formula for $*^2$,

$$\begin{aligned} * \delta \alpha &= (-1)^{np+n+1} * d(* \alpha) \\ &= (-1)^{np+n+1+(n-p+1)(p-1)} d(* \alpha) \\ &= (-1)^p d(* \alpha), \end{aligned}$$

because

$$np + n + 1 + (n - p + 1)(p - 1) \equiv p \pmod{2}.$$

Consequently,

$$\begin{aligned} (d\eta, \alpha)_{L^2} &= \int_X \eta \wedge *(\delta \alpha) \\ &= (\eta, \delta \alpha)_{L^2}. \end{aligned}$$

By the defining property and uniqueness of the formal adjoint,

$$d^* = \delta = (-1)^{np+n+1} * d *.$$

In normal orthonormal coordinates at a point q ,

$$d^* \left(f_{i_1 \dots i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p} \right)_q = - \sum_{r=1}^p (-1)^{r-1} \frac{\partial f_{i_1 \dots i_p}}{\partial x^{i_r}}(q) \cdot dx^{i_1} \wedge \dots \wedge \widehat{dx^{i_r}} \wedge \dots \wedge dx^{i_p}.$$

The Hodge Laplacian

[Definition] The **Hodge Laplacian** is

$$\Delta := dd^* + d^*d : \mathcal{A}^p(X) \longrightarrow \mathcal{A}^p(X).$$

A form α is **harmonic** if $\Delta\alpha = 0$. Write

$$\mathcal{H}^p(X) := \ker(\Delta : \mathcal{A}^p(X) \rightarrow \mathcal{A}^p(X)).$$

[Proposition] A form α is harmonic if and only if it is both closed and coclosed:

$$\Delta\alpha = 0 \iff d\alpha = 0 \text{ and } d^*\alpha = 0.$$

[Proof] By formal adjointness,

$$(\Delta\alpha, \alpha)_{L^2} = (dd^*\alpha, \alpha)_{L^2} + (d^*d\alpha, \alpha)_{L^2} = \|d^*\alpha\|_{L^2}^2 + \|d\alpha\|_{L^2}^2.$$

If $\Delta\alpha = 0$, both nonnegative terms vanish. The converse follows immediately from the definition of Δ .

[Proposition] The Hodge star intertwines the Laplacian:

$$*\Delta = \Delta*.$$

Moreover,

$$d\Delta = \Delta d, \quad d^*\Delta = \Delta d^*.$$

In oriented orthonormal coordinates centered at a point, the principal part of Δ is the scalar Laplacian acting on each coefficient. Thus, with the nonnegative sign convention used above,

$$\Delta \left(\sum_I f_I dx^{i_1} \wedge \dots \wedge dx^{i_p} \right) = - \sum_{I,j} \frac{\partial^2 f_I}{\partial (x^j)^2} dx^{i_1} \wedge \dots \wedge dx^{i_p} + \text{lower-order terms}.$$

This shows that Δ is elliptic.

[Proof] Combining

$$d^* = (-1)^{np+n+1} * d * \quad \text{and} \quad *^2 = (-1)^{p(n-p)} \text{Id}$$

gives, for $\alpha \in \mathcal{A}^p(X)$,

$$d^*(*\alpha) = (-1)^{p+1} * (d\alpha), \quad d(*\alpha) = (-1)^p * (d^*\alpha).$$

Applying these identities to the two terms of $\Delta = dd^* + d^*d$ yields

$$\Delta(*\alpha) = *(\Delta\alpha).$$

The remaining commutation relations follow algebraically from $d^2 = (d^*)^2 = 0$:

$$d\Delta = dd^*d = \Delta d, \quad d^*\Delta = d^*dd^* = \Delta d^*.$$

Finally, in normal orthonormal coordinates at a point, the first derivatives of the metric vanish. The principal parts of d and d^* are therefore their Euclidean ones, and expanding $dd^* + d^*d$ cancels all mixed second derivatives, leaving

$$-\sum_j \partial_j^2$$

on every coefficient. Its principal symbol is $|\xi|^2 \text{Id}$, which is invertible for $\xi \neq 0$; hence Δ is elliptic.

33 The Hodge theorem and Green operator

[Theorem] Hodge theorem. Let (X, g) be a compact oriented Riemannian manifold. Every de Rham cohomology class contains a unique harmonic representative. Equivalently, the natural map

$$\mathcal{H}^p(X) \longrightarrow H_{\text{dR}}^p(X; \mathbb{R}), \quad \alpha \longmapsto [\alpha],$$

is an isomorphism.

The analytic form of the theorem is the orthogonal decomposition

$$\mathcal{A}^p(X) = \mathcal{H}^p(X) \oplus \text{im } d \oplus \text{im } d^*.$$

Thus every $\alpha \in \mathcal{A}^p(X)$ admits a unique decomposition

$$\alpha = h + d\beta + d^*\gamma,$$

where $h \in \mathcal{H}^p(X)$, $\beta \in \mathcal{A}^{p-1}(X)$, and $\gamma \in \mathcal{A}^{p+1}(X)$; uniqueness refers to the three orthogonal summands.

[Definition] Let

$$H : \mathcal{A}^p(X) \longrightarrow \mathcal{H}^p(X)$$

be orthogonal projection. The **Green operator**

$$G : \mathcal{A}^p(X) \longrightarrow \mathcal{A}^p(X)$$

is zero on $\mathcal{H}^p(X)$ and is the inverse of Δ on $\mathcal{H}^p(X)^\perp$. Therefore

$$\Delta G = G\Delta = \text{Id} - H.$$

Since G commutes with d and d^* , the identity above becomes

$$\text{Id} = H + dd^*G + d^*dG.$$

Applied to a form α , this gives the canonical Hodge decomposition

$$\alpha = H\alpha + d(d^*G\alpha) + d^*(dG\alpha).$$

Day 8

34 Analytic construction of the Green operator

The proof of the Hodge theorem may be organized through Sobolev spaces and the spectral theorem. Let (X, g) be a compact oriented Riemannian manifold. On smooth p -forms define

$$\|\varphi\|_{W^{1,2}}^2 := (\varphi, \varphi)_{L^2} + (d\varphi, d\varphi)_{L^2} + (d^*\varphi, d^*\varphi)_{L^2}.$$

Its completion is the Sobolev space $W^{1,2}(X, \Lambda^p T^*X)$. The corresponding inner product is

$$\mathcal{D}(\varphi, \psi) = (\varphi, \psi)_{L^2} + (d\varphi, d\psi)_{L^2} + (d^*\varphi, d^*\psi)_{L^2}.$$

For smooth forms, formal adjointness gives

$$\mathcal{D}(\varphi, \psi) = (\varphi, (\text{Id} + \Delta)\psi)_{L^2}.$$

[Proposition] For every $\eta \in L^2(X, \Lambda^p T^*X)$ there is a unique $T\eta \in W^{1,2}(X, \Lambda^p T^*X)$ such that

$$\mathcal{D}(T\eta, \xi) = (\eta, \xi)_{L^2}$$

for every $\xi \in W^{1,2}(X, \Lambda^p T^*X)$. Viewed as an operator on L^2 , the map T is compact, self-adjoint, positive, and injective.

[Proof] The estimate

$$|(\eta, \xi)_{L^2}| \leq \|\eta\|_{L^2} \|\xi\|_{L^2} \leq \|\eta\|_{L^2} \|\xi\|_{W^{1,2}}$$

shows that $\xi \mapsto (\eta, \xi)_{L^2}$ is a bounded functional on $W^{1,2}$. The Riesz representation theorem therefore gives a unique $T\eta$.

The Rellich compactness theorem says that the inclusion

$$W^{1,2}(X, \Lambda^p T^*X) \hookrightarrow L^2(X, \Lambda^p T^*X)$$

is compact, so $T : L^2 \rightarrow L^2$ is compact. If $\eta, \zeta \in L^2$, then

$$(T\eta, \zeta)_{L^2} = \mathcal{D}(T\eta, T\zeta) = (\eta, T\zeta)_{L^2},$$

which proves self-adjointness. Also

$$(T\eta, \eta)_{L^2} = \mathcal{D}(T\eta, T\eta) \geq 0.$$

Finally, $T\eta = 0$ implies $(\eta, \xi)_{L^2} = 0$ for every smooth ξ ; hence $\eta = 0$.

The spectral theorem supplies an orthogonal decomposition

$$L^2(X, \Lambda^p T^*X) = \widehat{\bigoplus_{m=0}^{\infty} E_m},$$

where each E_m is finite-dimensional and

$$T|_{E_m} = \mu_m \text{Id}, \quad 1 \geq \mu_0 > \mu_1 > \cdots > 0, \quad \mu_m \longrightarrow 0.$$

We now identify these eigenspaces with eigenspaces of the Hodge Laplacian. If $u = T\eta$, then the defining equation says

$$\mathcal{D}(u, \xi) = (\eta, \xi)_{L^2}$$

for every test form ξ . For smooth forms, integration by parts gives

$$\mathcal{D}(u, \xi) = ((\text{Id} + \Delta)u, \xi)_{L^2}.$$

Consequently,

$$(\text{Id} + \Delta)u = \eta$$

in the weak, or distributional, sense. Since this holds for every η , we may write

$$(\text{Id} + \Delta)T = \text{Id}.$$

Take $e \in E_m$. The eigenvalue equation $Te = \mu_m e$ and the preceding weak identity give

$$e = (\text{Id} + \Delta)Te = \mu_m(\text{Id} + \Delta)e.$$

Because $\mu_m > 0$, this is equivalent to

$$\Delta e = \lambda_m e, \quad \lambda_m := \frac{1}{\mu_m} - 1.$$

Thus

$$\Delta|_{E_m} = \lambda_m \text{Id}.$$

Initially this equation is only distributional. Elliptic regularity first implies $e \in W^{2,2}$; applying the regularity theorem repeatedly gives $e \in W^{s,2}$ for every s . Sobolev embedding then gives $e \in C^\infty$. Hence every E_m consists of smooth Laplace eigenforms.

The relation between the two spectra is therefore

$$\mu_m = \frac{1}{1 + \lambda_m}, \quad \lambda_m = \frac{1}{\mu_m} - 1.$$

Since Δ is nonnegative, $\lambda_m \geq 0$ and hence $0 < \mu_m \leq 1$. Moreover,

$$\mu_m = 1 \iff \lambda_m = 0 \iff E_m \subset \ker \Delta.$$

Thus the eigenspace of T with eigenvalue 1 is precisely the harmonic space $\mathcal{H}^p(X)$.

On $\mathcal{H}^p(X)^\perp$, all Laplace eigenvalues are positive, so the inverse of Δ must act on E_m by multiplication by $1/\lambda_m$. In terms of μ_m this is

$$G|_{E_m} = \frac{1}{\lambda_m} \text{Id} = \frac{\mu_m}{1 - \mu_m} \text{Id} \quad \text{when } \mu_m < 1,$$

while $G = 0$ on $\mathcal{H}^p(X)$.

The zero eigenvalue of an elliptic operator on a compact manifold is isolated and has finite multiplicity. Therefore the first positive eigenvalue λ_1 satisfies $\lambda_1 > 0$. For $\eta = \sum_{\lambda_m > 0} \eta_m$ orthogonal to the harmonic space,

$$\|G\eta\|_{L^2}^2 = \sum_{\lambda_m > 0} \frac{1}{\lambda_m^2} \|\eta_m\|_{L^2}^2 \leq \frac{1}{\lambda_1^2} \|\eta\|_{L^2}^2.$$

This is the positive spectral gap argument proving that G is bounded on L^2 .

Finally, both identities can be checked on each spectral summand. If $e \in \mathcal{H}^p(X)$, then $Ge = \Delta e = 0$, while $(\text{Id} - H)e = 0$. If $e \in E_m$ with $\lambda_m > 0$, then

$$\Delta Ge = \Delta \left(\frac{1}{\lambda_m} e \right) = e, \quad G\Delta e = G(\lambda_m e) = e,$$

and $He = 0$. Hence, on the whole spectral decomposition,

$$\Delta G = G\Delta = \text{Id} - H.$$

35 Hermitian Hodge theory

Let (X, J, g) be a compact Hermitian manifold. The complexified exterior algebra decomposes orthogonally as

$$\mathcal{A}^r(X; \mathbb{C}) = \bigoplus_{p+q=r} \mathcal{A}^{p,q}(X).$$

Because the summands are orthogonal, ∂ and $\bar{\partial}$ have formal adjoints

$$\partial^* : \mathcal{A}^{p,q} \longrightarrow \mathcal{A}^{p-1,q}, \quad \bar{\partial}^* : \mathcal{A}^{p,q} \longrightarrow \mathcal{A}^{p,q-1}.$$

Define

$$\Delta_{\partial} = \partial\partial^* + \partial^*\partial, \quad \Delta_{\bar{\partial}} = \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}.$$

Both are elliptic, self-adjoint operators preserving bidegree. The Hodge theorem applied to this elliptic complex gives

$$H_{\bar{\partial}}^{p,q}(X) \cong \mathcal{H}_{\bar{\partial}}^{p,q}(X) := \ker \Delta_{\bar{\partial}} \cap \mathcal{A}^{p,q}(X).$$

By the Dolbeault theorem, the same space represents $H^q(X, \Omega_X^p)$.

36 Kähler identities

Now suppose that (X, J, g) is Kähler, with Kähler form

$$\omega(u, v) = g(Ju, v).$$

The **Lefschetz operator** and its adjoint are

$$L : \mathcal{A}^{p,q} \longrightarrow \mathcal{A}^{p+1,q+1}, \quad L\alpha = \omega \wedge \alpha, \quad \Lambda := L^*.$$

Define also

$$d^c := i(\bar{\partial} - \partial).$$

Then

$$dd^c = 2i\partial\bar{\partial}.$$

[Theorem] Kähler identities. On a Kähler manifold,

$$[\Lambda, \partial] = i\bar{\partial}^*, \quad [\Lambda, \bar{\partial}] = -i\partial^*.$$

Consequently,

$$\Delta_d = 2\Delta_{\partial} = 2\Delta_{\bar{\partial}}.$$

[Proof] The identities are tensorial and first order, so they may be checked at a point in holomorphic normal coordinates, where

$$\omega = \frac{i}{2} \sum_{j=1}^n dz^j \wedge d\bar{z}^j$$

and the first derivatives of the metric vanish. Write

$$\varepsilon_j(\alpha) = dz^j \wedge \alpha, \quad \bar{\varepsilon}_j(\alpha) = d\bar{z}^j \wedge \alpha,$$

and denote their metric adjoints by ε_j^* and $\bar{\varepsilon}_j^*$. Wedge multiplication and contraction satisfy the canonical anticommutation relations

$$\varepsilon_j^* \varepsilon_k + \varepsilon_k \varepsilon_j^* = \delta_{jk} \text{Id}, \quad \bar{\varepsilon}_j^* \bar{\varepsilon}_k + \bar{\varepsilon}_k \bar{\varepsilon}_j^* = \delta_{jk} \text{Id},$$

while operators of different holomorphic type anticommute.

At the chosen point, L is wedge multiplication by $\frac{i}{2} \sum_j dz^j \wedge d\bar{z}^j$, and $\Lambda = L^*$ is the corresponding sum of two contractions. Also, because the connection coefficients vanish at that point,

$$\partial = \sum_j \varepsilon_j \nabla_{\partial/\partial z^j}, \quad \bar{\partial} = \sum_j \bar{\varepsilon}_j \nabla_{\partial/\partial \bar{z}^j},$$

and their adjoints are obtained by replacing wedge multiplication with its adjoint and the covariant derivative with its formal adjoint. Expanding the commutators and using the anticommutation relations leaves only the terms with equal indices. With the normalization above, the result is

$$[\Lambda, \partial] = i\bar{\partial}^*, \quad [\Lambda, \bar{\partial}] = -i\partial^*.$$

Since the point was arbitrary, these identities hold globally.

It remains to derive the Laplacian identities. Expanding $d = \partial + \bar{\partial}$ and $d^* = \partial^* + \bar{\partial}^*$ gives

$$\Delta_d = \Delta_\partial + \Delta_{\bar{\partial}} + \partial\bar{\partial}^* + \bar{\partial}^*\partial + \bar{\partial}\partial^* + \partial^*\bar{\partial}.$$

The two mixed pairs vanish directly from the Kähler identities. Indeed, $\bar{\partial}^* = -i[\Lambda, \partial]$, so

$$\begin{aligned} \partial\bar{\partial}^* + \bar{\partial}^*\partial &= -i(\partial[\Lambda, \partial] + [\Lambda, \partial]\partial) \\ &= -i(\partial\Lambda\partial - \partial^2\Lambda + \Lambda\partial^2 - \partial\Lambda\partial) \\ &= 0. \end{aligned}$$

The same calculation with $\partial^* = i[\Lambda, \bar{\partial}]$ gives $\bar{\partial}\partial^* + \partial^*\bar{\partial} = 0$.

Finally, use $\partial^* = i[\Lambda, \bar{\partial}]$ to write

$$\begin{aligned} \Delta_\partial &= \partial\partial^* + \partial^*\partial \\ &= i(\partial\Lambda\bar{\partial} - \partial\bar{\partial}\Lambda + \Lambda\bar{\partial}\partial - \bar{\partial}\Lambda\partial). \end{aligned}$$

On the other hand, $\bar{\partial}^* = -i[\Lambda, \partial]$ gives

$$\begin{aligned} \Delta_{\bar{\partial}} &= \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial} \\ &= -i(\bar{\partial}\Lambda\partial - \bar{\partial}\partial\Lambda + \Lambda\partial\bar{\partial} - \partial\Lambda\bar{\partial}). \end{aligned}$$

Using $\partial\bar{\partial} = -\bar{\partial}\partial$ shows that these two expressions are equal. Therefore

$$\Delta_\partial = \Delta_{\bar{\partial}},$$

and the vanishing of the mixed terms now yields

$$\Delta_d = \Delta_\partial + \Delta_{\bar{\partial}} = 2\Delta_\partial = 2\Delta_{\bar{\partial}}.$$

[Corollary] On a compact Kähler manifold, the Hodge decomposition respects bidegree:

$$\mathcal{H}_d^r(X; \mathbb{C}) = \bigoplus_{p+q=r} \mathcal{H}_{\bar{\partial}}^{p,q}(X).$$

Consequently,

$$H_{\text{dR}}^r(X; \mathbb{C}) \cong \bigoplus_{p+q=r} H_{\bar{\partial}}^{p,q}(X).$$

[Proof] Since $\Delta_d = 2\Delta_{\bar{\partial}}$ and $\Delta_{\bar{\partial}}$ preserves bidegree, a form $\alpha = \sum_{p+q=r} \alpha^{p,q}$ is d -harmonic if and only if every $\alpha^{p,q}$ is $\bar{\partial}$ -harmonic. The cohomological decomposition then follows from the uniqueness of harmonic representatives.

[Corollary] Hodge symmetry. For a compact Kähler manifold,

$$h^{p,q}(X) = h^{q,p}(X).$$

[Proof] Complex conjugation sends a (p, q) -form to a (q, p) -form and commutes with the real operator Δ_d . By the Kähler identity it therefore gives an antilinear isomorphism

$$\mathcal{H}_{\bar{\partial}}^{p,q}(X) \longrightarrow \mathcal{H}_{\bar{\partial}}^{q,p}(X).$$

37 The dd^c -lemma

[Lemma] The dd^c -lemma. Let X be a compact Kähler manifold and let α be a d -closed form of pure type. If α is d -exact, then

$$\alpha \in \text{im}(dd^c) = \text{im}(i\partial\bar{\partial}).$$

In particular, if α is a real exact $(1, 1)$ -form, then there is a real-valued smooth function φ such that

$$\alpha = i\partial\bar{\partial}\varphi.$$

[Proof] Suppose that α has bidegree (p, q) . Since

$$d\alpha = \partial\alpha + \bar{\partial}\alpha = 0$$

and the two summands have the distinct bidegrees $(p+1, q)$ and $(p, q+1)$, each summand must vanish. Thus

$$\partial\alpha = 0, \quad \bar{\partial}\alpha = 0, \quad d^c\alpha = i(\bar{\partial} - \partial)\alpha = 0.$$

Apply the Hodge decomposition to α :

$$\alpha = H\alpha + dd^*G\alpha + d^*dG\alpha.$$

The first term is zero because α is exact. Indeed, if $\alpha = d\beta$ and h is harmonic, then $d^*h = 0$ and hence

$$(\alpha, h)_{L^2} = (d\beta, h)_{L^2} = (\beta, d^*h)_{L^2} = 0.$$

Thus α is orthogonal to the harmonic space and $H\alpha = 0$. Moreover, G commutes with d , so

$$dG\alpha = Gd\alpha = 0.$$

Consequently, the coexact term also vanishes and

$$\alpha = dd^*G\alpha.$$

We now convert d^* into a commutator with d^c . From the Kähler identities,

$$[\Lambda, \partial] = i\bar{\partial}^*, \quad [\Lambda, \bar{\partial}] = -i\partial^*.$$

Therefore

$$\begin{aligned} [\Lambda, d^c] &= i[\Lambda, \bar{\partial} - \partial] \\ &= i(-i\partial^* - i\bar{\partial}^*) \\ &= \partial^* + \bar{\partial}^* = d^*. \end{aligned}$$

The identity $\Delta_d = 2\Delta_\partial = 2\Delta_{\bar{\partial}}$ implies that d^c commutes with Δ_d . It therefore preserves every eigenspace of Δ_d , and hence commutes with the Green operator G . Using $d^c\alpha = 0$, we obtain

$$d^c G\alpha = G d^c \alpha = 0.$$

Substituting $d^* = [\Lambda, d^c]$ into the reduced Hodge decomposition now gives

$$\begin{aligned} \alpha &= d[\Lambda, d^c]G\alpha \\ &= d\Lambda d^c G\alpha - dd^c \Lambda G\alpha \\ &= -dd^c \Lambda G\alpha. \end{aligned}$$

Thus $\alpha = dd^c \beta$ with

$$\beta = -\Lambda G\alpha.$$

Since $dd^c = 2i\partial\bar{\partial}$, the images of dd^c and $i\partial\bar{\partial}$ coincide, proving the first assertion.

Finally, suppose that α is a real $(1, 1)$ -form. Both G and Λ preserve real forms, while G preserves bidegree and Λ lowers bidegree by $(1, 1)$. Hence $\Lambda G\alpha$ is a real-valued function. Setting

$$\varphi = -2\Lambda G\alpha$$

and using $dd^c = 2i\partial\bar{\partial}$ yields

$$\alpha = -dd^c \Lambda G\alpha = i\partial\bar{\partial}\varphi,$$

as required.

Day 9

38 Kodaira–Nakano vanishing

Let (X, ω) be a compact Kähler manifold. By the Dolbeault and Hodge theorems,

$$H^q(X, \Omega_X^p) \cong H_{\bar{\partial}}^{p,q}(X) \cong \mathcal{H}_{\bar{\partial}}^{p,q}(X).$$

Thus every Dolbeault class has a unique $\bar{\partial}$ -harmonic representative. This description is canonical, but the harmonic space is usually difficult to compute directly. The Bochner technique imposes a curvature condition which forces a harmonic representative to vanish.

[Definition] Let (L, h) be a Hermitian holomorphic line bundle. In a local holomorphic frame e , its Chern curvature is

$$\Theta_h = -\partial\bar{\partial} \log h(e, e).$$

The line bundle is **positive** if the real $(1, 1)$ -form $i\Theta_h$ is positive definite. Equivalently, $i\Theta_h$ is a Kähler form.

The Lefschetz operator is

$$L_\omega : A^{p,q}(X, L) \longrightarrow A^{p+1,q+1}(X, L), \quad L_\omega \alpha = \omega \wedge \alpha,$$

and its adjoint is $\Lambda = L_\omega^*$.

If $D = D^{1,0} + \bar{\partial}$ is the Chern connection of (L, h) , then the bundle-valued Kähler identities are

$$[\Lambda, \bar{\partial}] = -i(D^{1,0})^*, \quad [\Lambda, D^{1,0}] = i\bar{\partial}^*.$$

For the trivial line bundle these reduce to the ordinary Kähler identities, because $D^{1,0} = \partial$.

[Lemma] Bochner–Kodaira–Nakano identity. On L -valued forms,

$$\Delta_{\bar{\partial}} = \Delta_{D^{1,0}} + [i\Theta_h, \Lambda],$$

where

$$\Delta_{D^{1,0}} = D^{1,0}(D^{1,0})^* + (D^{1,0})^*D^{1,0}.$$

[Proof] The curvature of the Chern connection is

$$D^2 = D^{1,0}\bar{\partial} + \bar{\partial}D^{1,0} = \Theta_h.$$

Using the bundle-valued Kähler identities and expanding the commutators,

$$\begin{aligned} -i\Delta_{D^{1,0}} &= -i(D^{1,0}(D^{1,0})^* + (D^{1,0})^*D^{1,0}) \\ &= D^{1,0}[\Lambda, \bar{\partial}] + [\Lambda, \bar{\partial}]D^{1,0}, \end{aligned}$$

whereas

$$\begin{aligned} i\Delta_{\bar{\partial}} &= i(\bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}) \\ &= \bar{\partial}[\Lambda, D^{1,0}] + [\Lambda, D^{1,0}]\bar{\partial}. \end{aligned}$$

Subtracting and collecting terms gives

$$i(\Delta_{\bar{\partial}} - \Delta_{D^{1,0}}) = \Lambda(D^{1,0}\bar{\partial} + \bar{\partial}D^{1,0}) - (D^{1,0}\bar{\partial} + \bar{\partial}D^{1,0})\Lambda.$$

Since the operator in parentheses is Θ_h , division by i gives the claimed identity.

[Theorem] Kodaira–Akizuki–Nakano vanishing. Let (X, ω) be a compact Kähler manifold of complex dimension n , and let (L, h) be a positive Hermitian holomorphic line bundle. Then

$$H_{\bar{\partial}}^{p,q}(X, L) = 0 \quad \text{whenever } p + q > n.$$

In particular,

$$H^q(X, K_X \otimes L) = 0 \quad (q > 0),$$

where $K_X = \Lambda^n T^{*1,0}X = \Omega_X^n$ is the canonical bundle.

[Proof] Use the positive curvature form itself as the Kähler metric, so that

$$\omega = i\Theta_h.$$

At a point, take a unitary coframe for this metric. The elementary wedge–contraction calculation gives

$$[i\Theta_h, \Lambda]|_{A^{p,q}(X, L)} = (p + q - n) \text{Id}.$$

If $u \in \mathcal{H}_{\bar{\partial}}^{p,q}(X, L)$, the Bochner identity gives

$$\begin{aligned} 0 &= (\Delta_{\bar{\partial}} u, u)_{L^2} \\ &= (\Delta_{D^{1,0}} u, u)_{L^2} + ([i\Theta_h, \Lambda]u, u)_{L^2} \\ &= \|D^{1,0}u\|_{L^2}^2 + \|(D^{1,0})^*u\|_{L^2}^2 + (p+q-n)\|u\|_{L^2}^2. \end{aligned}$$

Every term is nonnegative, and the last coefficient is positive when $p+q > n$. Hence $u = 0$.

39 Kodaira embedding theorem

[Theorem] Kodaira embedding theorem. Let X be a compact complex manifold and let $L \rightarrow X$ be a positive holomorphic line bundle. For all sufficiently large m , a basis

$$s_0, \dots, s_N \in H^0(X, L^m), \quad N+1 = \dim_{\mathbb{C}} H^0(X, L^m),$$

defines a holomorphic embedding

$$\Phi_m : X \longrightarrow \mathbb{CP}^N, \quad x \longmapsto [s_0(x) : \dots : s_N(x)].$$

[Proof]

In a local frame e of L , write $s_j = f_j e^m$. Then

$$\Phi_m(x) = [f_0(x) : \dots : f_N(x)],$$

which is independent of the chosen frame because a change of frame multiplies all entries by the same nonzero function.

To prove that Φ_m is an embedding it is enough to establish the following three properties for all sufficiently large m :

(1) No base points. For every $p \in X$, some section of L^m is nonzero at p . Equivalently, the evaluation map

$$H^0(X, L^m) \longrightarrow L_p^m$$

is surjective.

(2) Separation of points. For $p \neq q$, the map

$$H^0(X, L^m) \longrightarrow L_p^m \oplus L_q^m$$

is surjective.

(3) Separation of tangent vectors. For each p , the first-jet map

$$H^0(X, L^m) \longrightarrow L_p^m \otimes_{\mathbb{C}} \mathcal{O}_{X,p}/\mathfrak{m}_p^2$$

is surjective.

The first property makes Φ_m everywhere defined, the second makes it injective, and the third makes its differential injective. Compactness then turns the resulting injective holomorphic immersion into an embedding.

The cohomological mechanism behind these properties is the exact sequence

$$0 \longrightarrow L^m \otimes \mathfrak{m}_p \longrightarrow L^m \longrightarrow L^m \otimes \mathcal{O}_X/\mathfrak{m}_p \longrightarrow 0.$$

Its long exact sequence shows that evaluation at p is surjective once

$$H^1(X, L^m \otimes \mathfrak{m}_p) = 0.$$

Likewise, the ideals $\mathfrak{m}_p \mathfrak{m}_q$ and $\mathfrak{m}_p^2$ control separation of two points and of first jets.

The ideal $\mathfrak{m}_p$ is not a line bundle, so Kodaira vanishing cannot be applied to it directly. Blow up X at p :

$$\pi : \hat{X} \longrightarrow X.$$

The map is an isomorphism away from p , and the exceptional set $E = \pi^{-1}(p)$ is a divisor. One has

$$\pi_* \mathcal{O}_{\hat{X}}(-E) = \mathfrak{m}_p, \quad R^i \pi_* \mathcal{O}_{\hat{X}}(-E) = 0 \quad (i > 0).$$

The projection formula therefore gives

$$H^1(X, L^m \otimes \mathfrak{m}_p) \cong H^1(\hat{X}, \pi^* L^m \otimes \mathcal{O}_{\hat{X}}(-E)).$$

There are two apparent difficulties. First, $\pi^* L$ is only semipositive: its curvature degenerates in directions tangent to E . Second, Kodaira vanishing applies to a bundle of the form $K_{\hat{X}} \otimes A$ with A positive. Both are resolved by the exceptional bundle. The bundle $\mathcal{O}_{\hat{X}}(-E)$ admits a metric whose curvature is positive in the exceptional directions near E . Moreover,

$$K_{\hat{X}} \cong \pi^* K_X \otimes \mathcal{O}_{\hat{X}}((n-1)E),$$

and hence

$$(\pi^* L^m \otimes \mathcal{O}(-E)) \otimes K_{\hat{X}}^{-1} \cong \pi^*(L^m \otimes K_X^{-1}) \otimes \mathcal{O}(-nE).$$

For m sufficiently large, the large positive horizontal curvature from L^m and the positive exceptional curvature together make this bundle positive. Kodaira vanishing then yields the required H^1 -vanishing. Blowing up two points and using higher powers of the exceptional ideal gives point and tangent separation. Compactness of X makes the choice of m uniform.

[Remark] Analytically, the same construction can be viewed as a $\bar{\partial}$ -problem. Choose a local frame e near p and a cutoff χ supported in that trivializing neighborhood. Solve

$$\bar{\partial} u = \bar{\partial}(\chi e^m).$$

Then $s = \chi e^m - u$ is holomorphic. A singular weight in the L^2 estimate forces $u(p) = 0$ and hence $s(p) = e^m(p) \neq 0$. Stronger singular weights prescribe first jets.

40 Hörmander L^2 estimates

[Theorem] Weighted estimate. Let (X, ω) be compact Kähler, let (L, h) be a Hermitian holomorphic line bundle, and let $\varphi \in L^1(X)$ be a weight such that, in the weak sense,

$$i\Theta_h + i\partial\bar{\partial}\varphi \geq c\omega$$

for a constant $c > 0$. If

$$g \in L^2(X, \Lambda^{n,1} T^* X \otimes L), \quad \bar{\partial} g = 0,$$

then there is

$$u \in L^2(X, \Lambda^{n,0} T^* X \otimes L)$$

such that $\bar{\partial}u = g$ and

$$\int_X |u|_h^2 e^{-\varphi} dV_\omega \leq \frac{1}{c} \int_X |g|_h^2 e^{-\varphi} dV_\omega.$$

The weight need not be C^2 . It is convenient to regard

$$\tilde{h} = h e^{-\varphi}$$

as a singular Hermitian metric; its curvature current is

$$i\Theta_{\tilde{h}} = i\Theta_h + i\partial\bar{\partial}\varphi.$$

For an L -valued $(0,1)$ -form, put

$$F = L \otimes K_X^{-1}.$$

Since $K_X \otimes F \cong L$, the preceding theorem solves $\bar{\partial}u = g$ for L -valued functions provided

$$i\Theta_h + \text{Ric}(\omega) + i\partial\bar{\partial}\varphi \geq c\omega.$$

This is the curvature assumption used in the analytic proof of the embedding theorem. Near a chosen point p , take holomorphic coordinates z and a singular weight of the form

$$\varphi(z) = n \log |z|^2$$

up to a smooth cutoff and a large smooth positive term. Then

$$e^{-\varphi} = |z|^{-2n}.$$

If u is continuous and

$$\int |u|^2 e^{-\varphi} < \infty,$$

the nonintegrability of $|z|^{-2n}$ forces $u(p) = 0$. This is exactly the vanishing needed in the cutoff construction above.

[Proof] The basic estimate obtained from the Bochner–Kodaira–Nakano identity is

$$\|\beta\|_{L^2}^2 \leq \frac{1}{c} (\|\bar{\partial}\beta\|_{L^2}^2 + \|\bar{\partial}^*\beta\|_{L^2}^2)$$

for smooth compactly supported L -valued $(n,1)$ -forms β , with all norms computed using the weighted metric $h e^{-\varphi}$.

Because $\bar{\partial}g = 0$, take a smooth test form β satisfying $\bar{\partial}\beta = 0$ and define on $\text{im } \bar{\partial}^*$ the functional

$$\ell_g(\bar{\partial}^*\beta) = (g, \beta)_{L^2}.$$

It is well defined on this subspace. Indeed, if $\bar{\partial}^*\beta = 0$ as well, the basic estimate forces $\beta = 0$. The basic estimate and Cauchy–Schwarz give

$$|\ell_g(\bar{\partial}^*\beta)| \leq \|g\|_{L^2} \|\beta\|_{L^2} \leq \frac{1}{\sqrt{c}} \|g\|_{L^2} \|\bar{\partial}^*\beta\|_{L^2}.$$

Thus ℓ_g extends by Hahn–Banach to a bounded functional on the ambient L^2 space. By the Riesz representation theorem, there is u such that

$$(u, \bar{\partial}^*\beta)_{L^2} = (g, \beta)_{L^2}$$

for every $\bar{\partial}$ -closed test form β . Since both $\bar{\partial}u$ and g are $\bar{\partial}$ -closed, testing on this subspace is enough to conclude that $\bar{\partial}u = g$ weakly. The operator norm gives

$$\|u\|_{L^2} \leq \frac{1}{\sqrt{c}} \|g\|_{L^2}.$$

Squaring proves the displayed estimate. Elliptic regularity makes u smooth wherever g , the metric, and the weight are smooth.